AP CALCULUS BC · EXAMINATION REPOSITORY

Unit 10

Infinite Sequences and Series

100-Question Priority-Starred Master Set

- Questions 1–90: Multiple Choice (Full Priority Ratings · Exactly 4 Questions Per Page)
- Questions 91–100: Free Response with Official AP Scoring Rubrics (9 Pts Each · 1 Per Page)
- Convergence Tests, Taylor/Maclaurin Series, Lagrange Error Bound & Power Series
- Complete 90-Question Answer Key & Step-by-Step Bulleted Solutions with Distractor Analysis

OFFICIAL AP CED ALIGNED · MEDIUM LEVEL · RIGOROUSLY VERIFIED

Examination Structure

This comprehensive paper contains 100 continuously numbered questions strictly covering Unit 10 (Infinite Sequences and Series), incorporating official AP-style convergence testing, error bounds, Taylor/Maclaurin series expansions, and multi-part FRQ rubrics to test conceptual and analytical calculus mastery.

⭐ Priority Rating System

★★★★★ VERY IMPORTANT	High-frequency core exam topics: Geometric Series, Ratio Test, Maclaurin Series for e^x, sin x, cos x, ln(1+x), 1/(1−x), Lagrange Error Bound, and Radius/Interval of Convergence.
★★★★ MANDATORY	Essential analytical skills: Integral Test & Remainder, p-Series Test, Comparison Tests (DCT/LCT), Alternating Series Test & Error Bound, Term-by-Term Differentiation/Integration, and Taylor Coefficients.
★★★ IMPORTANT	Foundational practice: nth-Term Test for Divergence, Partial sums sequences, Telescoping series, Repeating decimal representations, and Absolute vs. Conditional convergence classification.

- **Questions 1–90:** Multiple Choice, four choices, exactly one correct answer (Exactly 4 MCQs per page).
- **Questions 91–100:** Free Response Questions, 9 points each, with authentic AP multi-part rubrics (1 FRQ per page).
- **Visual Elements:** Step-by-step tables of values, derivative profiles, error bound intervals, and summation models.
- **Calculator Policy:** No calculator required (Closed-form analytical evaluation).
- **Answer Section:** Complete Answer Key grid, followed by step-by-step bulleted solutions and distractor breakdowns.

Topic Coverage (Unit 10.1 – 10.15)

10.1 Defining Convergent and Divergent Infinite Series
10.2 Working with Geometric Series & Partial Sums
10.3 The Integral Test for Convergence & Remainder
10.4 Harmonic Series and p-Series
10.5 Comparison Tests (Direct Comparison Test)
10.6 Limit Comparison Test (LCT)
10.7 Alternating Series Test for Convergence
10.8 Alternating Series Error Bound
10.9 Determining Absolute or Conditional Convergence
10.10 Ratio Test for Convergence
10.11 Taylor Polynomial Approximations of Functions
10.12 Lagrange Error Bound (Taylor's Remainder)
10.13 Radius and Interval of Convergence of Power Series
10.14 Finding Taylor or Maclaurin Series for a Function
10.15 Representing Functions as Power Series & Term-by-Term Calculus

Essential Tip: The Ratio Test yields L = 1 on all algebraic, rational, and p-series expressions (always inconclusive!). Use the Limit Comparison Test instead. For power series intervals of convergence, always test the endpoints individually—one may converge by AST while the other diverges by the harmonic series!

ITEM 01

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the sum of the infinite geometric series $\sum_{n=1}^{\infty} 4(-2/3)^n$?

A $-8/5$ **B** $8/5$

C $-12/5$ **D** The series diverges.

ITEM 02

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Which of the following statements is true about the infinite series $\sum_{n=1}^{\infty} (3n^2+5)/(5n^2-2n)$?

A The series converges to 3 / 5 by the nth-Term Test.

B The series diverges by the nth-Term Test for Divergence because lim_(n→∞) a_n = 3 / 5 ≠ 0.

C The nth-Term Test is inconclusive because lim_(n→∞) a_n = 3 / 5.

D The series converges by comparison with $\sum 1/n^2$.

ITEM 03

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the sum of the telescoping series $\sum_{n=1}^{\infty} 2/[n(n+2)]$?

A 2 **B** 1

C 3 / 2 **D** 3 / 4

ITEM 04

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The n-th partial sum of an infinite series $\sum_{n=1}^{\infty} a_n$ is given by S_n = (4n – 1) / (2n + 3). What is the value of the third term, a_3, of the series?

A 14 / 63 **B** 11 / 9

C 1 **D** 2 / 9

ITEM 05

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For what values of x does the infinite geometric series $\sum_{n=0}^{\infty} [(2x - 1) / 5]^n$ converge, and what is its sum in terms of x on that interval?

A Converges for $-2 < x < 3$ with sum $5 / (6 - 2x)$

B Converges for $-2 < x < 3$ with sum $5 / (4 - 2x)$

C Converges for $-3 < x < 2$ with sum $5 / (6 - 2x)$

D Converges for $-2 \le x \le 3$ with sum $1 / (6 - 2x)$

ITEM 06

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

Which of the following statements must be true for an infinite series $\sum_{n=1}^{\infty} a_n$?

A If $\lim_{n\to\infty} a_n = 0$, then the series $\sum_{n=1}^{\infty} a_n$ converges.

B If the series $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

C If the series $\sum_{n=1}^{\infty} a_n$ diverges, then $\lim_{n\to\infty} a_n \neq 0$.

D If $\lim_{n\to\infty} a_n \neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ converges to that limit.

ITEM 07

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

The repeating decimal 0.454545... can be represented as an infinite geometric series. What is its exact rational fraction representation?

A 45 / 100

B 9 / 20

C 5 / 11

D 45 / 999

ITEM 08

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The table below displays the terms a_n and partial sums S_n for an infinite series $\sum_{n=1}^{\infty} a_n$.

n	1	2	3	4	5
a_n	1/2	1/4	1/8	1/16	1/32
S_n	1/2	3/4	7/8	15/16	31/32

Based on the closed-form pattern of S_n, what is the exact sum of this infinite series?

A The series diverges.

B 2

C 63 / 64

D 1

ITEM 09

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the sum of the infinite series $\sum_{n=0}^{\infty} (2^n + 3^n) / 5^n$?

A 25 / 6
B 5 / 6
C 5
D 17 / 6

ITEM 10

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A ball is dropped from a height of 18 meters. Each time it strikes the ground, it rebounds to a height equal to 2 / 3 of its previous height. What is the total vertical distance traveled by the ball before coming to rest?

A 54 meters
B 90 meters
C 72 meters
D 108 meters

ITEM 11

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For what values of p does the infinite series $\sum_{n=1}^{\infty} 1 / n^{(2p - 3)}$ converge?

A p > 1
B p ≥ 2
C p > 2
D p > 4

ITEM 12

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

Which of the following sets of conditions must be satisfied by a function f(x) on the interval [1, ∞) in order to apply the Integral Test to the series $\sum_{n=1}^{\infty} f(n)$?

A f(x) must be alternating and decreasing for x ≥ 1.
B f(x) must be differentiable, positive, and increasing for x ≥ 1.
C f(x) must be continuous and bounded for x ≥ 1.
D f(x) must be continuous, positive, and decreasing for x ≥ 1.

ITEM 13

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Does the infinite series $\sum_{n=1}^{\infty} n / (n^2 + 1)$ converge or diverge, and which test provides the justification?

A Diverges by the Integral Test because $\int_1^{\infty} x / (x^2 + 1)\, dx = \infty$.

B Converges by the Integral Test because $\int_1^{\infty} x / (x^2 + 1)\, dx = \pi / 4$.

C Converges by the nth-Term Test because lim_(n→∞) n / (n² + 1) = 0.

D Diverges by the nth-Term Test because lim_(n→∞) n / (n² + 1) ≠ 0.

ITEM 14

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The convergent p-series $S = \sum_{n=1}^{\infty} 1 / n^3$ is approximated by its fourth partial sum $S_4 = \sum_{n=1}^{4} 1 / n^3$. According to the Integral Test Remainder Theorem, what is the maximum possible error in this approximation, $R_4 = S - S_4$?

A 1 / 16

B 1 / 32

C 1 / 64

D 1 / 48

ITEM 15

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Consider the two series: (I) $\sum_{n=2}^{\infty} 1 / [n \ln(n)]$ and (II) $\sum_{n=2}^{\infty} 1 / [n (\ln(n))^2]$. Which of the following is true?

A Both (I) and (II) diverge by the Integral Test.

B Both (I) and (II) converge by the Integral Test.

C (I) diverges and (II) converges by the Integral Test.

D (I) converges and (II) diverges by the Integral Test.

ITEM 16

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

For which real values of the constant k does the infinite series $\sum_{n=1}^{\infty} [1 / n^3 + k / 3^n]$ converge?

A |k| < 3 only

B |k| < 1 only

C k > 0 only

D All real numbers k

ITEM 17

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The table below lists values of the improper integral bound $\int_N^\infty 1/x^4\,dx$ for selected integer values of N.

N	2	3	4	5
$\int_N^\infty 1/x^4\,dx$	1/24	1/81	1/192	1/375

If the series $\sum_{n=1}^{\infty} 1/n^4$ is approximated by its N-th partial sum S_N, what is the smallest integer N guaranteed by the table to ensure the remainder R_N < 0.01?

A N = 4

B N = 3

C N = 5

D N = 2

ITEM 18

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

Which of the following explains why the infinite series $\sum_{n=1}^{\infty} 1/(3n+2)$ diverges?

A Its terms do not approach zero as n → ∞.

B It behaves like a constant multiple of the divergent harmonic series ∑ 1/n.

C It is a geometric series with common ratio r > 1.

D It is an alternating series that fails the second condition of the Alternating Series Test.

ITEM 19

MODERATE · SINGLE BEST CHOICE ★★★ IMPORTANT

If the series $\sum_{n=1}^{\infty} 1/n^{(p/2)}$ converges, which of the following must be true?

A p ≥ 2 and lim_(n→∞) 1/n^(p/2) ≠ 0

B p > 1 and lim_(n→∞) 1/n^(p/2) = 0

C p > 2 and lim_(n→∞) 1/n^(p/2) = 0

D 0 < p < 2

ITEM 20

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let f(x) be a continuous, positive, and decreasing function for x ≥ 1 such that the improper integral $\int_1^\infty f(x)\,dx$ converges to I. Which of the following inequalities correctly bounds the sum of the infinite series $S = \sum_{n=1}^{\infty} f(n)$?

A I/2 ≤ S ≤ I

B f(1) + I ≤ S ≤ 2I

C 0 ≤ S ≤ I

D I ≤ S ≤ f(1) + I

ITEM 21

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Which of the following tests and benchmark series can be used to show that $\sum_{n=1}^{\infty} 1/(n^3 + 5n + 2)$ converges?

A Direct Comparison Test with the convergent p-series $\sum 1/n^3$

B Direct Comparison Test with the divergent p-series $\sum 1/n$

C The nth-Term Test because $\lim_{n\to\infty} 1/(n^3 + 5n + 2) = 0$

D Geometric Series Test with common ratio $r = 1/3$

ITEM 22

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

Does the infinite series $\sum_{n=1}^{\infty} 1/(\sqrt{n} - 0.5)$ converge or diverge, and by which test?

A Converges by the Direct Comparison Test because $1/(\sqrt{n} - 0.5) < 1/\sqrt{n}$ and $\sum 1/\sqrt{n}$ converges.

B Diverges by the Direct Comparison Test because $1/(\sqrt{n} - 0.5) > 1/\sqrt{n}$ and $\sum 1/\sqrt{n}$ diverges.

C Converges by the p-Series Test because $p = 1/2 < 1$.

D Diverges by the nth-Term Test because $\lim_{n\to\infty} 1/(\sqrt{n} - 0.5) = \infty$.

ITEM 23

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Which test conclusively determines the convergence of the infinite series $\sum_{n=1}^{\infty} (2n^2 + 3)/\sqrt{(n^7 + 4n)}$?

A Converges by the Ratio Test with limit $L = 2/7$.

B Diverges by the Limit Comparison Test with the divergent harmonic series $\sum 1/n$.

C Converges by the Limit Comparison Test with the convergent p-series $\sum 1/n^{(3/2)}$.

D Diverges by the nth-Term Test because $\lim_{n\to\infty} a_n = 2$.

ITEM 24

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the convergence status of the infinite series $\sum_{n=1}^{\infty} (3 + \sin n)/n^2$?

A The Direct Comparison Test is inconclusive because $\sin n$ takes negative values.

B Diverges because $\sin n$ oscillates indefinitely between -1 and 1.

C Diverges by comparison with the harmonic series $\sum 1/n$.

D Converges by the Direct Comparison Test because $0 \le (3 + \sin n)/n^2 \le 4/n^2$.

ITEM 25

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let $\sum a_n$ and $\sum b_n$ be series with positive terms. Under which of the following conditions does the Limit Comparison Test guarantee that $\sum a_n$ converges?

A $\lim_{n\to\infty} a_n / b_n = 0$ and $\sum b_n$ converges.

B $\lim_{n\to\infty} a_n / b_n = 0$ and $\sum b_n$ diverges.

C $\lim_{n\to\infty} a_n / b_n = \infty$ and $\sum b_n$ converges.

D $\lim_{n\to\infty} a_n / b_n = 1$ and $\sum b_n$ diverges.

ITEM 26

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

To determine the convergence of the series $\sum_{n=2}^{\infty} (n+1)/(n^3-4)$, which of the following benchmark series should be used with the Limit Comparison Test, and what is the final conclusion?

A $b_n = 1/n$; the series diverges because $\lim_{n\to\infty} a_n / b_n = 1$ and $\sum 1/n$ diverges.

B $b_n = 1/n^2$; the series converges because $\lim_{n\to\infty} a_n / b_n = 1$ and $\sum 1/n^2$ converges.

C $b_n = 1/n^3$; the series converges because $\lim_{n\to\infty} a_n / b_n = 0$.

D $b_n = 1/n^2$; the series diverges because $n^3 - 4$ is smaller than n^3.

ITEM 27

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A student attempts to prove that the series $\sum_{n=1}^{\infty} 1/(n^2-0.5)$ converges by comparing it directly to the convergent p-series $\sum_{n=1}^{\infty} 1/n^2$. Why is this argument invalid?

A The terms $1/(n^2-0.5)$ are negative for all $n \geq 1$.

B The series $\sum 1/n^2$ actually diverges by the p-series test.

C The inequality $1/(n^2-0.5) > 1/n^2$ holds, so the terms are larger than the convergent series, making DCT inconclusive in that direction.

D Direct comparison is never permitted when the denominator contains a subtraction.

ITEM 28

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The table below displays the values of $a_n = 1/(2^n+1)$ alongside the terms of the geometric series $b_n = 1/2^n$.

n	1	2	3	4
$a_n = 1/(2^n+1)$	1/3	1/5	1/9	1/17
$b_n = 1/2^n$	1/2	1/4	1/8	1/16

Because $0 < a_n < b_n$ for all $n \geq 1$ and $\sum_{n=1}^{\infty} b_n = 1$, what does the Direct Comparison Test establish about the sum $S = \sum_{n=1}^{\infty} a_n$?

A The Direct Comparison Test is inconclusive because the ratio a_n / b_n is not constant.

B The series $\sum a_n$ diverges because $1/3 + 1/5 + 1/9 + \ldots$ grows without bound.

C The series $\sum a_n$ converges to exactly 1.

D The series $\sum a_n$ converges, and its sum satisfies $0 < S < 1$.

ITEM 29

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the convergence behavior of the series $\sum_{n=1}^{\infty} 1 / \sqrt{(n^3 + 2n)}$?

A Converges by the Limit Comparison Test with $\sum 1 / n^{(3/2)}$

B Diverges by the Limit Comparison Test with $\sum 1 / n^3$

C Diverges by the p-Series Test with $p = 3 / 2$

D Converges by the Ratio Test with limit $L = 0$

ITEM 30

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Which of the following justifies the convergence of the infinite series $\sum_{n=1}^{\infty} \arctan(n) / n^3$?

A nth-Term Test: $\lim_{n\to\infty} \arctan(n) / n^3 = 0$, so the series converges.

B Direct Comparison Test: $0 < \arctan(n) / n^3 < (\pi / 2) / n^3$, and $(\pi / 2) \sum 1 / n^3$ converges.

C Direct Comparison Test: $\arctan(n) / n^3 < 1 / n$, and $\sum 1 / n$ converges.

D Geometric Series Test: common ratio $r = \arctan(1) < 1$.

ITEM 31

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Which of the following justifies that the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} / \sqrt{n}$ converges?

A It converges absolutely by the Direct Comparison Test with $\sum 1 / n$.

B It converges by the p-Series Test because $p = 1 / 2$.

C It converges by the Alternating Series Test because $1 / \sqrt{n} > 0$, $1 / \sqrt{(n+1)} \le 1 / \sqrt{n}$, and $\lim_{n\to\infty} 1 / \sqrt{n} = 0$.

D It converges by the Ratio Test with limit $L = 0$.

ITEM 32

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The convergent alternating series $S = \sum_{n=1}^{\infty} (-1)^{n+1} / n^2$ is approximated by its fifth partial sum $S_5 = 1 - 1/4 + 1/9 - 1/16 + 1/25$. According to the Alternating Series Error Bound, what is the maximum possible value of the error $|S - S_5|$?

A $1 / 6$

B $1 / 25$

C $1 / 49$

D $1 / 36$

ITEM 33

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

Let $S = \sum_{n=1}^{\infty} (-1)^{n+1} / (2n + 1) = 1/3 - 1/5 + 1/7 - 1/9 + \ldots$ be a convergent alternating series. Is the fourth partial sum $S_4 = 1/3 - 1/5 + 1/7 - 1/9$ an overestimate or an underestimate of the true sum S, and why?

A An underestimate, because the first omitted term is positive (+1 / 11).

B An overestimate, because the last included term is negative (−1 / 9).

C An overestimate, because $S_4 > S$ for all alternating series.

D Neither, because S_4 equals the exact sum.

ITEM 34

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The alternating series $S = \sum_{n=1}^{\infty} (-1)^n / n^3$ is to be approximated by its N-th partial sum S_N such that the error $|S - S_N| < 0.001$. What is the smallest positive integer N guaranteed by the Alternating Series Error Bound to satisfy this condition?

A 9

B 10

C 11

D 100

ITEM 35

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

Which of the following tests correctly establishes the convergence of the infinite series $\sum_{n=1}^{\infty} \cos(\pi n) / n$?

A Divergence Test, because $\cos(\pi n)$ fluctuates infinitely.

B p-Series Test with p = 1.

C Alternating Series Test, because $\cos(\pi n) = (-1)^n$ and 1 / n decreases monotonically to 0.

D Geometric Series Test with common ratio r = −1.

ITEM 36

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

Why does the alternating series $\sum_{n=1}^{\infty} (-1)^n (3n + 1) / (2n - 1)$ diverge?

A It converges to 0 by the Alternating Series Test.

B It diverges by the Alternating Series Test because the terms are not positive.

C It diverges by the p-Series Test because p = 1.

D It diverges by the nth-Term Test because lim_(n→∞) a_n does not exist (the terms oscillate near ±3/2).

ITEM 37

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The table below lists the first four partial sums S_n of an alternating series $S = \sum_{n=1}^{\infty} (-1)^{n+1} b_n$ where b_n decreases to 0.

n	1	2	3	4
S_n	1.000	0.500	0.833	0.583

If the fifth term magnitude is $b_5 = 0.200$, which of the following intervals is guaranteed to contain the exact infinite sum S?

A [0.583, 0.783]

B [0.383, 0.583]

C [0.500, 1.000]

D [0.783, 0.833]

ITEM 38

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The series $S = \sum_{n=1}^{\infty} (-1)^{n+1} / (2n)!$ is approximated by its second partial sum $S_2 = 1/2! - 1/4! = 1/2 - 1/24 = 11/24$. What is the maximum error in this approximation according to the Alternating Series Error Bound?

A 1/120

B 1/720

C 1/24

D 1/5040

ITEM 39

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Why is the condition that $b_{(n+1)} \le b_n$ (terms are eventually decreasing in magnitude) strictly required in the Alternating Series Test, even if $\lim_{n\to\infty} b_n = 0$?

A Without monotonicity, the series would automatically become a geometric series.

B Monotonicity is not required; any series with $\lim b_n = 0$ and alternating signs must converge.

C Without monotonicity, the positive terms can sum to infinity while the negative terms sum to a finite value, causing the series to diverge.

D The condition only ensures that the sum of the series is positive.

ITEM 40

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The Maclaurin series for cos x evaluated at x = 0.5 is given by $\cos(0.5) = 1 - (0.5)^2/2! + (0.5)^4/4! - (0.5)^6/6! + \ldots$. If cos(0.5) is approximated by its first two terms $1 - (0.5)^2/2 = 0.875$, what is the maximum error in this approximation guaranteed by the Alternating Series Error Bound?

A 1/768

B 1/192

C 1/48

D 1/384

ITEM 41

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Which of the following describes the convergence of the infinite series $\sum_{n=1}^{\infty} (-1)^{n+1} / n$?

A Conditionally convergent

B Absolutely convergent

C Divergent

D Inconclusive by the Alternating Series Test

ITEM 42

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The alternating series $\sum_{n=1}^{\infty} (-1)^n / n^3$ is:

A Conditionally convergent because the signs alternate.

B Absolutely convergent because $\sum 1 / n^3$ converges by the p-series test.

C Divergent by the nth-Term test.

D Conditionally convergent because $p = 3 > 1$.

ITEM 43

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What does the Ratio Test conclude about the infinite series $\sum_{n=1}^{\infty} 3^n / n!$?

A Inconclusive because $L = 1$.

B Diverges because $L = \lim_{n\to\infty} |a_{n+1} / a_n| = 3 > 1$.

C Converges absolutely because $L = \lim_{n\to\infty} |a_{n+1} / a_n| = 0 < 1$.

D Diverges because 3^n grows exponentially.

ITEM 44

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the result of applying the Ratio Test to the series $\sum_{n=1}^{\infty} 5^n / n^3$?

A The series converges because $p = 3 > 1$.

B The series converges because $L = 1 / 5 < 1$.

C The test is inconclusive because $L = 1$.

D The series diverges because $L = 5 > 1$.

ITEM 45

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

For which of the following infinite series is the Ratio Test inconclusive (yielding $L = 1$)?

A $\sum_{n=1}^{\infty} (n^2 + 1) / (n^4 + 3)$

B $\sum_{n=1}^{\infty} 2^n / n!$

C $\sum_{n=1}^{\infty} (3 / 4)^n$

D $\sum_{n=1}^{\infty} n! / 10^n$

ITEM 46

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Does the alternating series $\sum_{n=1}^{\infty} (-1)^n / \sqrt{(n^2 + 1)}$ converge absolutely, converge conditionally, or diverge?

A Converges absolutely

B Converges conditionally

C Diverges by the nth-Term Test

D Diverges by the Integral Test

ITEM 47

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What does the Ratio Test establish for the series $\sum_{n=1}^{\infty} (n!)^2 / (2n)!$?

A Converges absolutely because $L = 1 / 2 < 1$

B Diverges because $L = 4 > 1$

C Converges absolutely because $L = 1 / 4 < 1$

D Inconclusive because $L = 1$

ITEM 48

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The table below records the ratio of consecutive terms $|a_{n+1} / a_n|$ for an infinite series $\sum_{n=1}^{\infty} a_n$ at increasing values of n.

n	1	2	5	10	50	100
$\lvert a_{n+1} / a_n \rvert$	0.850	0.780	0.720	0.708	0.701	0.700

Based on the limiting trend evident in the table, what conclusion does the Ratio Test provide regarding $\sum_{n=1}^{\infty} a_n$?

A The series converges conditionally to 0.700.

B The series diverges because the terms are non-zero.

C The Ratio Test is inconclusive because the ratio decreases.

D The series converges absolutely because $\lim_{n \to \infty} |a_{n+1} / a_n| = 0.700 < 1$.

ITEM 49

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

If an infinite series $\sum_{n=1}^{\infty} a_n$ is known to be absolutely convergent, which of the following must be true?

A The series $\sum_{n=1}^{\infty} a_n$ converges, and $\left|\sum_{n=1}^{\infty} a_n\right| \le \sum_{n=1}^{\infty} |a_n|$.

B The series $\sum_{n=1}^{\infty} a_n$ converges conditionally.

C The series $\sum_{n=1}^{\infty} a_n$ diverges.

D $\lim_{n\to\infty} a_n = 1$.

ITEM 50

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Consider the three infinite series:
(I) $\sum_{n=1}^{\infty} (-1)^n / n^2$
(II) $\sum_{n=1}^{\infty} (-1)^n / n$
(III) $\sum_{n=1}^{\infty} (-1)^n n / (n+1)$
Which of these series converges conditionally?

A (I) and (II) only

B (II) only

C (I) only

D (III) only

ITEM 51

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the third-degree Taylor polynomial for $f(x) = \ln(x)$ centered at $c = 1$?

A $(x-1) + (1/2)(x-1)^2 + (1/3)(x-1)^3$

B $(x-1) - (x-1)^2 + 2(x-1)^3$

C $(x-1) - (1/2)(x-1)^2 + (1/3)(x-1)^3$

D $1 + (x-1) - (1/2)(x-1)^2 + (1/3)(x-1)^3$

ITEM 52

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The fourth-degree Maclaurin polynomial for a function $f(x)$ is given by $P_4(x) = 3 - 2x + 5x^2 - 4x^3 + 7x^4$. What is the value of $f'''(0)$?

A 42

B −4

C −12

D −24

ITEM 53

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let f(x) be a function with third-degree Taylor polynomial $P_3(x)$ centered at c = 2. It is known that $|f^{(4)}(t)| \le 12$ for all t in the interval [2, 2.5]. What is the maximum possible value of the error $|f(2.5) - P_3(2.5)|$ guaranteed by the Lagrange Error Bound?

A 1/32 **B** 1/16

C 1/64 **D** 1/8

ITEM 54

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

Using the second-degree Maclaurin polynomial for $f(x) = e^x$, what is the approximation for $e^{-0.2}$?

A 0.80 **B** 0.82

C 0.84 **D** 0.78

ITEM 55

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For the approximation of $e^{-0.2}$ by $P_2(-0.2)$ in the previous question, what is the upper bound on the error $|e^{-0.2} - P_2(-0.2)|$ provided by the Lagrange Error Bound on [−0.2, 0]?

A 1/300 **B** 1/150

C 1/750 **D** 1/125

ITEM 56

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the second-degree Taylor polynomial for $f(x) = \sqrt{x}$ centered at c = 4?

A $2 + (1/4)(x-4) + (1/64)(x-4)^2$ **B** $2 + (1/4)(x-4) - (1/32)(x-4)^2$

C $2 + (1/2)(x-4) - (1/16)(x-4)^2$ **D** $2 + (1/4)(x-4) - (1/64)(x-4)^2$

ITEM 57

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A function f has derivatives of all orders. Selected values of f and its derivatives at x = 2 are given in the table below.

f(2)	f'(2)	f''(2)	f'''(2)
5	−3	8	−12

What is the third-degree Taylor polynomial for f centered at x = 2?

A $5 - 3(x-2) + 4(x-2)^2 - 2(x-2)^3$

B $5 - 3(x-2) + 8(x-2)^2 - 12(x-2)^3$

C $5 - 3x + 4x^2 - 2x^3$

D $5 - 3(x-2) + 4(x-2)^2 - 6(x-2)^3$

ITEM 58

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A fourth-degree Maclaurin polynomial $P_4(x)$ is used to approximate a function f(x) at x = 0.5. If the fifth derivative of f satisfies $|f^{(5)}(x)| \le 48$ for all x in [0, 0.5], what is the upper bound on the error guaranteed by the Lagrange Error Bound?

A 1/40

B 1/80

C 1/120

D 1/160

ITEM 59

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The second-degree Taylor polynomial for a function f about x = 3 is $P_2(x) = 7 - 4(x-3)^2$. Which of the following statements about f at x = 3 must be true?

A f has a point of inflection at x = 3.

B f has a relative minimum at x = 3 because f'(3) = 0 and f''(3) = −4 < 0.

C f has a relative maximum at x = 3 because f'(3) = 0 and f''(3) = −8 < 0.

D f is increasing at x = 3.

ITEM 60

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

When approximating the value of an alternating Taylor series that satisfies the conditions of the Alternating Series Test, what is the primary advantage of using the Alternating Series Error Bound over the Lagrange Error Bound?

A The Alternating Series Error Bound gives the exact error rather than an upper bound.

B The Alternating Series Error Bound always produces a strictly smaller bound than Lagrange.

C The Lagrange Error Bound cannot be used if a series has alternating signs.

D The Alternating Series Error Bound only requires evaluating the magnitude of the first omitted term, without maximizing an unknown derivative.

ITEM 61

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the interval of convergence of the power series $\sum_{n=1}^{\infty} (x-2)^n / (n \cdot 3^n)$?

A $[-1, 5)$
B $(-1, 5]$
C $(-1, 5)$
D $[-1, 5]$

ITEM 62

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the interval of convergence of the power series $\sum_{n=1}^{\infty} (x+1)^n / (n^2 \cdot 4^n)$?

A $(-5, 3)$
B $[-5, 3]$
C $[-5, 3)$
D $(-5, 3]$

ITEM 63

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the interval of convergence of the geometric power series $\sum_{n=0}^{\infty} (2x-4)^n / 5^n$?

A $(-1/2, 9/2]$
B $[-1/2, 9/2]$
C $(-1/2, 9/2)$
D $(-5, 5)$

ITEM 64

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the radius of convergence of the power series $\sum_{n=0}^{\infty} n!\,(x-3)^n$?

A $R = 3$
B $R = \infty$ (converges for all real x)
C $R = 1$
D $R = 0$ (converges only at $x = 3$)

ITEM 65

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the interval of convergence of the power series $\sum_{n=0}^{\infty} (x-1)^n / n!$?

A $(-\infty, \infty)$
B $[0, 2]$
C $(0, 2)$
D $x = 1$ only

ITEM 66

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A power series $\sum_{n=0}^{\infty} a_n (x-4)^n$ is known to converge at $x = 7$ and diverge at $x = 0$. Which of the following statements must be true?

A The series diverges at $x = 2$ and converges at $x = 9$.
B The series converges at $x = 2$ and diverges at $x = 9$.
C The series converges at $x = 9$.
D The radius of convergence must be $R = 3$.

ITEM 67

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

Consider the power series $\sum_{n=1}^{\infty} (x-5)^n / (\sqrt{n} \cdot 2^n)$. The table below summarizes the convergence behavior when testing candidate points.

Test Point	Series Form	Behavior
Interior $x \in (3, 7)$	Ratio Test $L = \lvert x-5 \rvert / 2 < 1$	Converges Absolutely
$x = 7$	$\sum 1 / \sqrt{n}$	Diverges ($p = 1/2$)
$x = 3$	$\sum (-1)^n / \sqrt{n}$	Converges (AST)

Based on this verified analysis, what is the full interval of convergence?

A $(3, 7)$
B $(3, 7]$
C $[3, 7)$
D $[3, 7]$

ITEM 68

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the interval of convergence of the power series $\sum_{n=1}^{\infty} x^{(2n)} / (n \cdot 4^n)$?

A $(-4, 4)$
B $[-2, 2]$
C $[-2, 2)$
D $(-2, 2)$

ITEM 69

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

Suppose $f(x) = \sum_{n=0}^{\infty} a_n (x-2)^n$ has radius of convergence $R = 5$. What is the radius of convergence of its derivative $f'(x) = \sum_{n=1}^{\infty} n a_n (x-2)^{n-1}$?

A $R = 5$
B $R = 4$
C $R = 25$
D $R = 0$

ITEM 70

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the interval of convergence of the power series $\sum_{n=1}^{\infty} (-1)^n (x+3)^n / n$?

A $[-4, -2)$
B $(-4, -2]$
C $(-4, -2)$
D $[-4, -2]$

ITEM 71

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the Maclaurin series for $f(x) = x^2 e^{-3x}$?

A $\sum_{n=0}^{\infty} [(-1)^n 3^n x^n] / n!$
B $\sum_{n=0}^{\infty} [3^n x^{(n+2)}] / n!$
C $\sum_{n=0}^{\infty} [(-1)^n 3^n x^{(n+2)}] / n!$
D $\sum_{n=0}^{\infty} [(-3)^n x^{(2n)}] / n!$

ITEM 72

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the coefficient of x^5 in the Maclaurin series for $f(x) = \sin(2x)$?

A $-4/15$
B $32/5$
C $2/15$
D $4/15$

ITEM 73

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the Maclaurin series for $f(x) = x / (1 + 4x^2)$, and on what interval does it converge?

A $\sum_{n=0}^{\infty} (-1)^n 4^n x^{(2n+1)}$ for $|x| < 1/2$

B $\sum_{n=0}^{\infty} 4^n x^{(2n+1)}$ for $|x| < 1/2$

C $\sum_{n=0}^{\infty} (-1)^n 4^n x^{(2n)}$ for $|x| < 2$

D $\sum_{n=0}^{\infty} (-1)^n 2^n x^{(2n+1)}$ for $|x| < 1/4$

ITEM 74

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the Taylor series for $f(x) = \sin x$ centered at $c = \pi / 2$?

A $\sum_{n=0}^{\infty} [(-1)^n (x - \pi/2)^{(2n+1)}] / (2n+1)!$

B $\sum_{n=0}^{\infty} [(-1)^n (x - \pi/2)^{(2n)}] / (2n)!$

C $\sum_{n=0}^{\infty} (x - \pi/2)^n / n!$

D $\sum_{n=0}^{\infty} [(-1)^{n+1} (x - \pi/2)^{(2n)}] / (2n)!$

ITEM 75

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

What are the first three non-zero terms of the Maclaurin series for $f(x) = \cos(x^2)$?

A $x^2 - x^6/6 + x^{10}/120$

B $1 - x^2/2 + x^4/24$

C $1 - x^4/2 + x^8/24$

D $1 + x^4/2 + x^8/24$

ITEM 76

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the exact sum of the infinite series $\sum_{n=0}^{\infty} [(-1)^n (\pi / 3)^{(2n)}] / (2n)!$?

A 1

B $\sqrt{3}/2$

C 0

D $1/2$

ITEM 77

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The table below shows the coefficients a_n of the Maclaurin series $\sum_{n=0}^{\infty} a_n x^n$ for an elementary function f(x).

n	0	1	2	3	4	5
a_n	1	0	−1/2	0	1/24	0

Which of the following functions matches this Maclaurin series?

A $f(x) = \cos x$

B $f(x) = \sin x$

C $f(x) = e^x$

D $f(x) = 1/(1+x^2)$

ITEM 78

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the exact numerical sum of the infinite series $\sum_{n=0}^{\infty} 2^n / n! = 1 + 2 + 4/2! + 8/3! + \ldots$?

A $2e$

B e^2

C $e/2$

D $1/(1-2)$

ITEM 79

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the exact sum of the alternating harmonic series $1 - 1/2 + 1/3 - 1/4 + 1/5 - \ldots$?

A $\ln(1/2)$

B $1/2$

C $\ln 2$

D $\pi/4$

ITEM 80

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What are the first three non-zero terms of the Maclaurin series for $f(x) = e^x \cos x$?

A $1 + x^2 - x^4/24$

B $1 + x - x^2/2$

C $1 + x + x^2/2$

D $1 + x - x^3/3$

ITEM 81

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the approximation for the non-elementary definite integral $\int_0^1 e^{-x^2}\,dx$ obtained by integrating the first three non-zero terms of its Maclaurin series?

A 23 / 30
B 7 / 10
C 13 / 15
D 11 / 15

ITEM 82

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the Maclaurin series for the function $F(x) = \int_0^x [\sin(t)/t]\,dt$?

A $\sum_{n=0}^{\infty} [(-1)^n x^{(2n+1)}]/(2n+1)!$
B $\sum_{n=0}^{\infty} [(-1)^n x^{(2n+1)}]/[(2n+1)(2n+1)!]$
C $\sum_{n=0}^{\infty} [(-1)^n x^{(2n)}]/(2n)!$
D $\sum_{n=0}^{\infty} [(-1)^n x^{(2n+2)}]/[(2n+2)!]$

ITEM 83

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Starting from the geometric series $1/(1-x) = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$, what is the closed-form rational function represented by the series $\sum_{n=1}^{\infty} n x^{n-1}$?

A $x/(1-x)^2$
B $1/(1-x)^3$
C $1/(1-x)^2$
D $\ln(1-x)$

ITEM 84

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the value of the limit $\lim_{x\to 0} (e^x - 1 - x)/x^2$?

A Does not exist
B 1
C 0
D 1/2

ITEM 85

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the value of the limit $\lim_{x\to 0} (x - \sin x) / x^3$?

A 1/6
B −1/6
C 1/3
D 0

ITEM 86

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

By integrating the geometric series $1 / (1 + t^2) = \sum_{n=0}^{\infty} (-1)^n t^{(2n)}$ from t = 0 to t = 1, which of the following famous infinite series formulas for π is established?

A $\pi = 2[1 - 1/3 + 1/5 - 1/7 + \dots]$
B $\pi = 4[1 - 1/3 + 1/5 - 1/7 + 1/9 - \dots]$
C $\pi = 4[1 + 1/3 + 1/5 + 1/7 + \dots]$
D $\pi = 1 - 1/2 + 1/3 - 1/4 + \dots$

ITEM 87

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The table below shows the first two terms in the series expansion of the definite integral $\int_0^{1/2} \cos(x^2)\,dx$ obtained by integrating $\cos(x^2) = 1 - x^4/2 + x^8/24 - \dots$.

Term Index n	Series Term	Integral Contribution	Running Sum
n = 0	1	$\int_0^{1/2} 1\,dx = 1/2$	0.500000
n = 1	$-x^4/2$	$\int_0^{1/2} (-x^4/2)\,dx = -1/160$	0.493750

According to the Alternating Series Error Bound, what is the maximum possible error in approximating this integral by the running sum 0.493750?

A 1/160
B 1/55296
C 1/110592
D 1/4608

ITEM 88

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the exact sum of the infinite series $\sum_{n=1}^{\infty} n / 2^n = 1/2 + 2/4 + 3/8 + 4/16 + \dots$?

A 3/2
B 1
C 4
D 2

ITEM 89

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A function f has Maclaurin series $f(x) = \sum_{n=0}^{\infty} [(-1)^n x^{(2n+1)}] / [(2n+1) \cdot 3^n]$. What is the value of $f^{(7)}(0)$?

A −80 / 3

B −5040 / 189

C −1 / 189

D 80 / 3

ITEM 90

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The definite integral $\int_0^1 [(1 - \cos x) / x] \, dx$ is approximated using the first non-zero term of its Maclaurin series. What is the value of this single-term approximation, and what is the error bound guaranteed by the Alternating Series Error Bound?

A Approximation = 1 / 2 with error at most 1 / 24

B Approximation = 1 / 4 with error at most 1 / 96

C Approximation = 1 / 4 with error at most 1 / 24

D Approximation = 1 / 6 with error at most 1 / 120

FRQ 1: Maclaurin Series from a Differential Equation

★★★★★ VERY IMPORTANT | 9 Points

Consider the differential equation $dy/dx = 2x + y$ with the initial condition $y(0) = 1$. Let $f(x)$ be the particular solution to this differential equation.

(a) [2 Points] Find $f'(0)$ and $f''(0)$.

(b) [2 Points] Find $f'''(0)$ and write the third-degree Maclaurin polynomial $P_3(x)$ for $f(x)$.

(c) [3 Points] Use the third-degree Maclaurin polynomial $P_3(x)$ to approximate $f(0.2)$. Show the calculation.

(d) [2 Points] Determine whether the graph of f has a relative minimum, relative maximum, or neither at $x = 0$. Justify your answer.

Official AP Scoring Directive: Show all supporting work. Clearly identify convergence tests, series formulas, error bounds, and derivative evaluations. Express answers in simplest exact analytical form.

FRQ 2: Power Series Interval of Convergence & Endpoint Analysis

★★★★★ VERY IMPORTANT

9 Points

Consider the function f defined by the power series $f(x) = \sum_{n=1}^{\infty} [(-1)^n (x-2)^n] / [n \cdot 4^n]$.

(a) [3 Points] Find the radius of convergence of the power series for f.

(b) [2 Points] Determine whether the series converges at the endpoint $x = 6$. Justify your answer.

(c) [2 Points] Determine whether the series converges at the endpoint $x = -2$. Justify your answer.

(d) [2 Points] Write the complete interval of convergence for f(x).

Official AP Scoring Directive: Show all supporting work. Clearly identify convergence tests, series formulas, error bounds, and derivative evaluations. Express answers in simplest exact analytical form.

FRQ 3: Taylor Polynomial & Lagrange Error Bound with Derivative Table

★★★★★ VERY IMPORTANT

9 Points

Let f be a function having derivatives of all orders for all real numbers. Selected values of f and its first three derivatives at x = 0 are given in the table below.

x	f(x)	f'(x)	f''(x)	f'''(x)
0	3	−2	6	−18

It is also known that the fourth derivative of f satisfies $|f^{(4)}(x)| \le 48$ for all x in the interval $[0, 0.5]$.

(a) [3 Points] Write the third-degree Maclaurin polynomial $P_3(x)$ for f.

(b) [2 Points] Use $P_3(x)$ to approximate $f(0.5)$.

(c) [2 Points] Use the Lagrange Error Bound to show that the approximation found in part (b) differs from $f(0.5)$ by at most $1/8$.

(d) [2 Points] Determine whether f has a relative extremum or point of inflection at x = 0. Justify your answer.

Official AP Scoring Directive: Show all supporting work. Clearly identify convergence tests, series formulas, error bounds, and derivative evaluations. Express answers in simplest exact analytical form.

FRQ 4: Alternating Series Convergence & Remainder Estimation

★★★★★ VERY IMPORTANT

9 Points

Consider the infinite alternating series $S = \sum_{n=1}^{\infty} (-1)^{n+1} / (2n - 1) = 1 - 1/3 + 1/5 - 1/7 + 1/9 - \dots$.

(a) [3 Points] Show that the series converges using the Alternating Series Test.

(b) [2 Points] Determine whether the series converges absolutely or conditionally. Justify your answer.

(c) [2 Points] Let S_3 be the third partial sum of the series. Show that $|S - S_3| < 0.15$. Is S_3 an overestimate or an underestimate of S?

(d) [2 Points] Find the smallest number of terms N such that the partial sum S_N approximates S with an error less than 0.01.

Official AP Scoring Directive: Show all supporting work. Clearly identify convergence tests, series formulas, error bounds, and derivative evaluations. Express answers in simplest exact analytical form.

FRQ 5: Term-by-Term Integration of a Maclaurin Series

★★★★★ VERY IMPORTANT | 9 Points

Consider the function $f(x) = (1 - e^{-x}) / x$ for $x \neq 0$ and $f(0) = 1$.

(a) [3 Points] Write the first four non-zero terms and the general term of the Maclaurin series for $f(x)$.

(b) [2 Points] Let $G(x) = \int_0^x f(t)\,dt$. Write the first four non-zero terms of the Maclaurin series for $G(x)$.

(c) [2 Points] Use the first two terms of the Maclaurin series for $G(x)$ to approximate $G(1)$.

(d) [2 Points] Explain why the approximation in part (c) differs from $G(1)$ by less than $1 / 18$.

Official AP Scoring Directive: Show all supporting work. Clearly identify convergence tests, series formulas, error bounds, and derivative evaluations. Express answers in simplest exact analytical form.

FRQ 6: Geometric Power Series, Derivative Series & Numerical Sums

★★★★★ VERY IMPORTANT

9 Points

Let $g(x) = 1 / (1 + 3x)$ for all x in the interval of convergence of its Maclaurin series.

(a) [2 Points] Write the Maclaurin series for $g(x)$ using summation notation, and state its interval of convergence.

(b) [3 Points] Differentiate the Maclaurin series for $g(x)$ to find a power series representation for $g'(x) = -3 / (1 + 3x)^2$.

(c) [2 Points] Use the series from part (b) to find a power series representation for $h(x) = x / (1 + 3x)^2$.

(d) [2 Points] Find the exact numerical sum of the infinite series $\sum_{n=1}^{\infty} n(-1)^{n+1}(3/4)^n$.

Official AP Scoring Directive: Show all supporting work. Clearly identify convergence tests, series formulas, error bounds, and derivative evaluations. Express answers in simplest exact analytical form.

FRQ 7: Taylor Series from Function Values & Derivative Table

★★★★★ VERY IMPORTANT

9 Points

A function f has derivatives of all orders for all real numbers. The table below gives values of f and its first four derivatives at x = 3.

f(3)	f'(3)	f''(3)	f'''(3)	f^(4)(3)
2	−4	10	−36	144

(a) [3 Points] Write the fourth-degree Taylor polynomial $P_4(x)$ for f centered at x = 3.

(b) [2 Points] Write the second-degree Taylor polynomial for the derivative f'(x) centered at x = 3.

(c) [2 Points] Use $P_4(x)$ to approximate f(3.1).

(d) [2 Points] Find the value of the limit $\lim_{x\to 3} [\, f(x) - 2 + 4(x-3) \,] / (x-3)^2$.

Official AP Scoring Directive: Show all supporting work. Clearly identify convergence tests, series formulas, error bounds, and derivative evaluations. Express answers in simplest exact analytical form.

FRQ 8: Ratio Test, Radius of Convergence & Function Identification

★★★★★ VERY IMPORTANT

9 Points

Consider the power series defined by $f(x) = \sum_{n=0}^{\infty} [(-1)^n x^{(2n)}] / 9^n$.

(a) [3 Points] Find the radius and interval of convergence of the power series.

(b) [2 Points] Find the closed-form elementary function represented by f(x) on its interval of convergence.

(c) [2 Points] Find the exact value of $f(\sqrt{3})$.

(d) [2 Points] Find a power series representation for $\int f(x)\,dx$ centered at $x = 0$.

Official AP Scoring Directive: Show all supporting work. Clearly identify convergence tests, series formulas, error bounds, and derivative evaluations. Express answers in simplest exact analytical form.

FRQ 9: Applied Convergence Modeling — Medical Drug Accumulation

★★★★★ VERY IMPORTANT

9 Points

A patient receives a daily injection of 150 mg of a therapeutic drug. Over each 24-hour period between injections, the patient's body eliminates 60% of the drug present, so that 40% remains just before the next injection.

(a) [2 Points] Find the amount of drug in the patient's body immediately after the second injection and immediately after the third injection.

(b) [3 Points] Write an expression for the amount of drug, D_n, present in the body immediately after the n-th injection.

(c) [2 Points] Assuming the injections continue indefinitely, find the limiting maximum amount of drug in the patient's body immediately after an injection.

(d) [2 Points] Find the limiting minimum amount of drug in the patient's body just before an injection in the long run.

Official AP Scoring Directive: Show all supporting work. Clearly identify convergence tests, series formulas, error bounds, and derivative evaluations. Express answers in simplest exact analytical form.

FRQ 10: Comprehensive Unit 10 Capstone Synthesis

★★★★★ VERY IMPORTANT 9 Points

The function f has Maclaurin series given by $f(x) = \sum_{n=0}^{\infty} [(-1)^n x^{(2n+1)}] / [(2n+1)! \cdot 2^n] = x - x^3/12 + x^5/480 - \ldots$ for all real numbers x.

(a) [2 Points] Find $f'(0)$ and $f'''(0)$.

(b) [3 Points] Find the radius of convergence of the Maclaurin series for f.

(c) [2 Points] Let $P_3(x) = x - x^3/12$ be the third-degree Maclaurin polynomial for f. Use the Alternating Series Error Bound to show that $|f(1) - P_3(1)| < 1/400$.

(d) [2 Points] Evaluate the limit $\lim_{x \to 0} [f(x) - x] / x^3$.

Official AP Scoring Directive: Show all supporting work. Clearly identify convergence tests, series formulas, error bounds, and derivative evaluations. Express answers in simplest exact analytical form.

End of Question Paper

You have concluded Questions 1 through 100 of the Unit 10 Master Examination Repository.

> **Important Study Strategy:** Carefully compare your solutions against the detailed answer keys and explanations that follow. Pay special attention to why incorrect choices were selected as distractors and how full credit is awarded on official AP rubrics.

Answer Section Contents

- **Answer Key Table:** Complete quick-reference grid of letter choices for Questions 1–90.
- **MCQ Detailed Solutions:** Step-by-step bulleted solutions for Questions 1–90 with full distractor error analysis.
- **FRQ Model Solutions:** Full analytical solutions for Questions 91–100 with official AP 9-point rubric point breakdowns.
- **Formula Appendix:** Summary of high-yield Unit 10 convergence tests, Maclaurin series formulas, and Lagrange error theorems.

Multiple Choice Answer Key

Verified answer keys with priority ratings. Questions 91–100 are free response (90 points total).

Q01	Q02	Q03	Q04	Q05	Q06	Q07	Q08	Q09	Q10
A ★★★★★	B ★★★★★	C ★★★★	D ★★★★	A ★★★★★	B ★★★★	C ★★★	D ★★★★	A ★★★★	B ★★★★★
Q11	**Q12**	**Q13**	**Q14**	**Q15**	**Q16**	**Q17**	**Q18**	**Q19**	**Q20**
C ★★★★★	D ★★★★	A ★★★★★	B ★★★★	C ★★★★★	D ★★★★	A ★★★★	B ★★★★	C ★★★	D ★★★★★
Q21	**Q22**	**Q23**	**Q24**	**Q25**	**Q26**	**Q27**	**Q28**	**Q29**	**Q30**
A ★★★★★	B ★★★★	C ★★★★★	D ★★★★	A ★★★★★	B ★★★★	C ★★★★★	D ★★★★	A ★★★★	B ★★★★★
Q31	**Q32**	**Q33**	**Q34**	**Q35**	**Q36**	**Q37**	**Q38**	**Q39**	**Q40**
C ★★★★★	D ★★★★★	A ★★★★	B ★★★★★	C ★★★★	D ★★★★	A ★★★★	B ★★★★	C ★★★★★	D ★★★★★
Q41	**Q42**	**Q43**	**Q44**	**Q45**	**Q46**	**Q47**	**Q48**	**Q49**	**Q50**
A ★★★★★	B ★★★★★	C ★★★★★	D ★★★★	A ★★★★	B ★★★★★	C ★★★★★	D ★★★★	A ★★★★	B ★★★★★
Q51	**Q52**	**Q53**	**Q54**	**Q55**	**Q56**	**Q57**	**Q58**	**Q59**	**Q60**
C ★★★★★	D ★★★★★	A ★★★★★	B ★★★★	C ★★★★★	D ★★★★	A ★★★★★	B ★★★★★	C ★★★★	D ★★★★
Q61	**Q62**	**Q63**	**Q64**	**Q65**	**Q66**	**Q67**	**Q68**	**Q69**	**Q70**
A ★★★★★	B ★★★★★	C ★★★★	D ★★★★	A ★★★★	B ★★★★★	C ★★★★	D ★★★★★	A ★★★★	B ★★★★
Q71	**Q72**	**Q73**	**Q74**	**Q75**	**Q76**	**Q77**	**Q78**	**Q79**	**Q80**
C ★★★★★	D ★★★★★	A ★★★★	B ★★★★★	C ★★★★	D ★★★★★	A ★★★★	B ★★★★	C ★★★★	D ★★★★★
Q81	**Q82**	**Q83**	**Q84**	**Q85**	**Q86**	**Q87**	**Q88**	**Q89**	**Q90**
A ★★★★★	B ★★★★	C ★★★★★	D ★★★★	A ★★★★	B ★★★★★	C ★★★★★	D ★★★★★	A ★★★★★	B ★★★★

Scoring Summary: Each multiple choice question is worth 1 point (90 points total). Each free response question is worth 9 points (90 points total). Total Exam Value: 180 Points.

Question 01 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) −8 / 5

STEP-BY-STEP SOLUTION

- **Identify the First Term a_1:**
- • Substitute n = 1 into the general term:
- • $a_1 = 4(-2/3)^1 = 4(-2/3) = -8/3$.
- **Identify the Common Ratio r:**
- • The common ratio between successive terms is $r = -2/3$.
- **Verify the Convergence Condition:**
- • A geometric series converges if and only if $|r| < 1$.
- • Here, $|r| = |-2/3| = 2/3 < 1$, so the series converges.
- **Apply the Geometric Series Sum Formula:**
- • $\text{Sum} = a_1/(1-r) = (-8/3)/[1-(-2/3)] = (-8/3)/[1+2/3] = (-8/3)/(5/3) = -8/5$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sign error in substituting n = 1, falsely evaluating the first term as +8/3.

✗ **(C)** Uses a = 4 (the coefficient at n = 0) instead of the actual first term at n = 1.

✗ **(D)** Incorrectly concludes divergence because the common ratio is negative.

Question 02 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) The series diverges by the nth-Term Test for Divergence because $\lim_{n\to\infty} a_n = 3/5 \neq 0$.

STEP-BY-STEP SOLUTION

- **State the nth-Term Test for Divergence:**
- • If $\lim_{n\to\infty} a_n$ does not exist or if $\lim_{n\to\infty} a_n \neq 0$, then the infinite series $\sum a_n$ must diverge.
- **Evaluate the Limit of the Sequence of Terms:**
- • $\lim_{n\to\infty} a_n = \lim_{n\to\infty} (3n^2+5)/(5n^2-2n)$.
- • Divide numerator and denominator by n^2:
- • $\lim_{n\to\infty} [3+5/n^2]/[5-2/n] = (3+0)/(5-0) = 3/5$.
- **Form the Conclusion:**
- • Because $\lim_{n\to\infty} a_n = 3/5 \neq 0$, the terms do not approach zero.
- • Therefore, the series **diverges** by the nth-Term Test.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses the limit of the individual terms $\lim a_n$ with the sum of the infinite series.

✗ **(C)** Falsely claims the nth-Term test is inconclusive when the limit is non-zero (it is only inconclusive when $\lim a_n = 0$).

✗ **(D)** Incorrectly assumes the series converges by comparing dominant terms without checking the limit.

Question 03 Analysis

★★★★ MANDATORY

Correct Answer: (C) 3 / 2

STEP-BY-STEP SOLUTION

- **Decompose into Partial Fractions:**
- • $2/[n(n+2)] = A/n + B/(n+2)$.
- • $2 = A(n+2) + Bn$.
- • For n = 0: $2 = 2A \Rightarrow A = 1$.
- • For n = −2: $2 = -2B \Rightarrow B = -1$.
- • Thus, $a_n = 1/n - 1/(n+2)$.
- **Write Out the First Several Terms to Observe Cancellation:**
- • n = 1: 1 − 1/3
- • n = 2: 1/2 − 1/4
- • n = 3: 1/3 − 1/5
- • n = 4: 1/4 − 1/6
- • Notice that −1/3 cancels with +1/3, and −1/4 cancels with +1/4.
- **Analyze the N-th Partial Sum S_N:**
- • $S_N = (1+1/2) - 1/(N+1) - 1/(N+2) = 3/2 - 1/(N+1) - 1/(N+2)$.
- **Take the Limit as $N \to \infty$:**
- • $\text{Sum} = \lim_{N\to\infty} S_N = 3/2 - 0 - 0 = 3/2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds all numerators without accounting for partial fraction decomposition.

✗ **(B)** Forgets the second surviving term 1/2, keeping only the first term 1.

✗ **(D)** Divides by an extra factor of 2 or makes an arithmetic error in 1 + 1/2.

Question 04 Analysis

★★★★ MANDATORY

Correct Answer: (D) 2 / 9

STEP-BY-STEP SOLUTION

- **Understand the Relation Between Terms and Partial Sums:**
- • By definition, $S_n = a_1 + a_2 + \ldots + a_n$.
- • The third term a_3 is obtained by subtracting S_2 from S_3: $a_3 = S_3 - S_2$.
- **Compute S_3:**
- • $S_3 = [4(3) - 1] / [2(3) + 3] = [12 - 1] / [6 + 3] = 11 / 9$.
- **Compute S_2:**
- • $S_2 = [4(2) - 1] / [2(2) + 3] = [8 - 1] / [4 + 3] = 7 / 7 = 1$.
- **Calculate a_3:**
- • $a_3 = S_3 - S_2 = 11 / 9 - 1 = 11 / 9 - 9 / 9 = 2 / 9$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Computes $S_3 - S_1$ instead of $S_3 - S_2$.

✗ **(B)** Reports the partial sum $S_3 = 11 / 9$ instead of the individual term a_3.

✗ **(C)** Reports the partial sum $S_2 = 1$.

Question 05 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) Converges for $-2 < x < 3$ with sum $5 / (6 - 2x)$

STEP-BY-STEP SOLUTION

- **Identify the Ratio and First Term:**
- • The series is geometric with first term $a = [(2x - 1) / 5]^0 = 1$.
- • The common ratio is $r = (2x - 1) / 5$.
- **Find the Interval of Convergence:**
- • A geometric series converges if and only if $|r| < 1$:
- • $|(2x - 1) / 5| < 1 \Rightarrow -1 < (2x - 1) / 5 < 1$.
- • Multiply by 5: $-5 < 2x - 1 < 5$.
- • Add 1: $-4 < 2x < 6$.
- • Divide by 2: $-2 < x < 3$.
- **Compute the Sum Function:**
- • Sum $= a / (1 - r) = 1 / [1 - (2x - 1) / 5] = 1 / [(5 - (2x - 1)) / 5] = 1 / [(6 - 2x) / 5] = 5 / (6 - 2x)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sign error in distributing the negative: $5 - 2x - 1 = 4 - 2x$ instead of $6 - 2x$.

✗ **(C)** Reverses the interval endpoints: $-3 < x < 2$.

Question 06 Analysis

★★★★ MANDATORY

Correct Answer: (B) If the series $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

STEP-BY-STEP SOLUTION

- **Recall the Fundamental Theorem of Series Convergence:**
- • If $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.
- • This is the contrapositive of the nth-Term Test for Divergence, making it universally **true**.
- **Analyze the Counterexamples to the Other Options:**
- • Option B is false: The harmonic series $\sum 1/n$ has $\lim 1/n = 0$, yet it diverges.
- • Option C is false: The harmonic series diverges even though $\lim 1/n = 0$.
- • Option D is false: If $\lim a_n \neq 0$, the series diverges, never converges.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Converse error: confusing a necessary condition with a sufficient condition (counterexample: harmonic series).

✗ **(C)** Inverse error: assuming divergence guarantees non-zero limit of terms.

✗ **(D)** Gross conceptual error confusing the limit of terms with series convergence.

Question 07 Analysis

★★★ IMPORTANT

Correct Answer: (C) 5 / 11

STEP-BY-STEP SOLUTION

- **Deconstruct into Geometric Components:**
- • 0.454545... = 0.45 + 0.0045 + 0.000045 + ...
- • = 45/100 + 45/10000 + 45/1000000 + ...
- **Identify First Term and Common Ratio:**
- • First term $a_1 = 45 / 100$.
- • Common ratio $r = (45/10000) / (45/100) = 1 / 100$.
- **Apply Geometric Sum Formula:**
- • Sum $= a_1 / (1 - r) = (45 / 100) / [1 - 1 / 100] = (45 / 100) / (99 / 100) = 45 / 99$.
- **Reduce the Fraction:**
- • Divide numerator and denominator by 9:
- • $45 / 99 = 5 / 11$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Treats it as a terminating decimal 45 / 100.

✗ **(B)** Takes only the first term 45 / 100 = 9 / 20.

Question 08 Analysis

★★★★ MANDATORY

Correct Answer: (D) 1

STEP-BY-STEP SOLUTION

- **Find the General Formula for S_n:**
- • Looking at the partial sums:
- • $S_1 = 1/2 = 1 - 1/2$
- • $S_2 = 3/4 = 1 - 1/4$
- • $S_3 = 7/8 = 1 - 1/8$
- • $S_4 = 15/16 = 1 - 1/16$
- • $S_5 = 31/32 = 1 - 1/32$
- • In general: $S_n = 1 - (1/2)^n$.
- **Determine Infinite Sum via the Limit of Partial Sums:**
- • By definition, the sum S of the infinite series is the limit of its sequence of partial sums:
- • $S = \lim_{n\to\infty} S_n = \lim_{n\to\infty} [1 - (1/2)^n]$.
- • Since $\lim_{n\to\infty} (1/2)^n = 0$: $S = 1 - 0 = \mathbf{1}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Falsely claims divergence despite the sequence of partial sums being bounded and strictly increasing.

Question 09 Analysis

★★★★ MANDATORY

Correct Answer: (A) 25 / 6

STEP-BY-STEP SOLUTION

- **Split the Series into Two Geometric Series:**
- • $(2^n + 3^n) / 5^n = (2/5)^n + (3/5)^n$.
- • $\sum_{n=0}^{\infty} (2^n + 3^n) / 5^n = \sum_{n=0}^{\infty} (2/5)^n + \sum_{n=0}^{\infty} (3/5)^n$.
- **Evaluate the First Geometric Series:**
- • First term $a_1 = (2/5)^0 = 1$, ratio $r_1 = 2/5$ ($|r_1| < 1$).
- • $\text{Sum}_1 = 1 / [1 - 2/5] = 1 / (3/5) = 5 / 3$.
- **Evaluate the Second Geometric Series:**
- • First term $a_2 = (3/5)^0 = 1$, ratio $r_2 = 3/5$ ($|r_2| < 1$).
- • $\text{Sum}_2 = 1 / [1 - 3/5] = 1 / (2/5) = 5 / 2$.
- **Combine the Sums:**
- • Total Sum $= 5 / 3 + 5 / 2 = 10 / 6 + 15 / 6 = 25 / 6$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Multiplies denominators or makes an arithmetic mistake in common denominator addition.

✗ **(C)** Falsely adds ratios (2/5 + 3/5 = 1) and claims sum is 5.

✗ **(D)** Evaluates the series starting at n = 1 instead of n = 0.

Question 10 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 90 meters

STEP-BY-STEP SOLUTION

- **Deconstruct the Motion:**
- • The ball drops 18 meters initially.
- • On every subsequent bounce, the ball travels BOTH upward and downward (twice the rebound height).
- • First rebound: up 18(2/3) = 12 m, down 12 m (total 24 m).
- • Second rebound: up 12(2/3) = 8 m, down 8 m (total 16 m).
- **Set Up the Infinite Series:**
- • Total Distance = $18 + 2 \cdot [\,18(2/3) + 18(2/3)^2 + 18(2/3)^3 + \ldots\,]$.
- • Total Distance = $18 + 2 \cdot [\,12 / (1 - 2/3)\,]$.
- **Evaluate the Sum:**
- • $12 / (1 - 2/3) = 12 / (1/3) = 36$ meters.
- • Total Distance = $18 + 2(36) = 18 + 72 =$ **90 meters**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Calculates $18 / (1 - 2/3) = 54$ meters, forgetting that the ball travels both up and down on every bounce.

✗ **(C)** Computes only the rebound distance $2(36) = 72$ meters, omitting the initial 18-meter drop.

Question 11 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) p > 2

STEP-BY-STEP SOLUTION

- **Recall the p-Series Convergence Criterion:**
- • A series of the form $\sum_{n=1}^{\infty} 1 / n^k$ converges if and only if the exponent $k > 1$.
- • If $k \le 1$, the series diverges.
- **Set Up the Inequality for the Exponent:**
- • Here, the power of n in the denominator is $k = 2p - 3$.
- • To converge: $2p - 3 > 1$.
- **Solve for p:**
- • $2p > 4 \Rightarrow p > 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sets $p > 1$ instead of setting the entire exponent $2p - 3 > 1$.

✗ **(B)** Includes the endpoint $p = 2$ (where $2(2) - 3 = 1$, giving the divergent harmonic series).

✗ **(D)** Arithmetic error in solving $2p - 3 > 1$ (e.g. $2p > 8$).

Question 12 Analysis

★★★★ MANDATORY

Correct Answer: (D) f(x) must be continuous, positive, and decreasing for x ≥ 1.

STEP-BY-STEP SOLUTION

- **State the Hypotheses of the Integral Test:**
- • Let f(x) be a function defined on $[1, \infty)$ such that $a_n = f(n)$.
- • The Integral Test requires three essential hypotheses on $[1, \infty)$:
- 1. **Continuous** on $[1, \infty)$
- 2. **Positive** $(f(x) > 0)$
- 3. **Decreasing** $(f'(x) \le 0)$
- • Under these conditions, the series $\sum_{n=1}^{\infty} f(n)$ and the improper integral $\int_1^{\infty} f(x)\,dx$ either both converge or both diverge.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses the Integral Test with the Alternating Series Test.

✗ **(B)** Requires increasing, which violates the test and guarantees divergence by the nth-Term test.

✗ **(C)** Omits the essential decreasing and positive requirements.

Question 13 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) Diverges by the Integral Test because $\int_1^\infty x/(x^2+1)\,dx = \infty$.

STEP-BY-STEP SOLUTION

- **Verify Conditions for $f(x) = x/(x^2+1)$:**
- • For $x \geq 1$: $f(x)$ is continuous and positive.
- • $f'(x) = [(x^2+1)(1) - x(2x)]/(x^2+1)^2 = (1-x^2)/(x^2+1)^2 \leq 0$ for $x \geq 1$, so f is decreasing.
- **Evaluate the Improper Integral:**
- • $\int_1^\infty x/(x^2+1)\,dx = \lim_{b\to\infty} \int_1^b x/(x^2+1)\,dx$.
- • Using substitution $u = x^2+1$ ($du = 2x\,dx \Rightarrow x\,dx = du/2$):
- • $= \lim_{b\to\infty} [(1/2)\ln(x^2+1)]_1^b = (1/2)\lim_{b\to\infty}[\ln(b^2+1) - \ln(2)] = \infty$.
- **Conclusion:**
- • Because the improper integral diverges, the series $\sum_{n=1}^\infty n/(n^2+1)$ **diverges** by the Integral Test.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Incorrectly calculates a finite value for a divergent logarithmic integral.
✗ **(C)** nth-Term test is inconclusive when $\lim a_n = 0$; it cannot prove convergence.
✗ **(D)** Calculates the limit of terms incorrectly (the limit is 0, not non-zero).

Question 14 Analysis

★★★★ MANDATORY

Correct Answer: (B) 1 / 32

STEP-BY-STEP SOLUTION

- **State the Remainder Bound for the Integral Test:**
- • If $f(x)$ is continuous, positive, and decreasing for $x \geq N$, the remainder $R_N = S - S_N$ satisfies:
- • $R_N \leq \int_N^\infty f(x)\,dx$.
- **Set Up the Integral for $N = 4$ and $f(x) = x^{-3}$:**
- • $R_4 \leq \int_4^\infty x^{-3}\,dx = \lim_{b\to\infty}[-(1/2)x^{-2}]_4^b$.
- **Evaluate the Limit:**
- • $= \lim_{b\to\infty}[-1/(2b^2) - (-1/(2(4)^2))]$
- • $= 0 + 1/(2\cdot 16) = 1/32$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the 1/2 coefficient from integration: 1 / 16.
✗ **(C)** Uses an extra power of 2 or cubes the denominator incorrectly.
✗ **(D)** Integrates with an incorrect power rule factor: $1/(3\cdot 16) = 1/48$.

Question 15 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) (I) diverges and (II) converges by the Integral Test.

STEP-BY-STEP SOLUTION

- **Analyze Series (I): $f(x) = 1/[x\ln(x)]$:**
- • Both x and $\ln(x)$ are positive and increasing for $x \geq 2$, so $f(x)$ is continuous, positive, and decreasing.
- • $\int_2^\infty 1/[x\ln(x)]\,dx$: Let $u = \ln(x) \Rightarrow du = dx/x$.
- • $= \int_{\ln 2}^\infty (1/u)\,du = [\ln(u)]_{(\ln 2)}^\infty = \lim_{u\to\infty}\ln(u) - \ln(\ln 2) = \infty$.
- • Integral diverges $\Rightarrow$ Series (I) **diverges**.
- **Analyze Series (II): $g(x) = 1/[x(\ln(x))^2]$:**
- • Continuous, positive, and decreasing for $x \geq 2$.
- • $\int_2^\infty 1/[x(\ln(x))^2]\,dx$: Let $u = \ln(x) \Rightarrow du = dx/x$.
- • $= \int_{\ln 2}^\infty u^{-2}\,du = [-1/u]_{(\ln 2)}^\infty = 0 - (-1/\ln 2) = 1/\ln 2 < \infty$.
- • Integral converges $\Rightarrow$ Series (II) **converges**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Falsely assumes any series with $\ln(n)$ in the denominator diverges.
✗ **(B)** Falsely concludes Series (I) converges because its terms approach 0 quickly.
✗ **(D)** Reverses the convergence behavior of the two series.

Question 16 Analysis

★★★★ MANDATORY

Correct Answer: (D) All real numbers k

STEP-BY-STEP SOLUTION

- **Use the Linearity of Convergent Series:**
- • The series can be split into two components: $\sum_{n=1}^{\infty} 1/n^3 + k \sum_{n=1}^{\infty} (1/3)^n$.
- **Analyze the First Component:**
- • $\sum 1/n^3$ is a p-series with $p = 3 > 1$, so it converges to a finite sum.
- **Analyze the Second Component:**
- • $\sum (1/3)^n$ is a geometric series with common ratio $r = 1/3$.
- • Because $|r| = 1/3 < 1$, the geometric series converges to a finite sum for any constant multiple k.
- **Conclusion:**
- • The sum of two convergent series multiplied by any scalar k is always convergent.
- • Therefore, the series converges for **all real numbers k**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misinterprets 3^n in the denominator as requiring $|k| < 3$.

✗ **(B)** Confuses the scalar k with the common ratio r of a geometric series.

✗ **(C)** Falsely assumes k must be strictly positive.

Question 17 Analysis

★★★★ MANDATORY

Correct Answer: (A) N = 4

STEP-BY-STEP SOLUTION

- **Apply the Integral Test Remainder Theorem:**
- • $R_N \leq \int_N^{\infty} 1/x^4\,dx$.
- • To ensure $R_N < 0.01 = 1/100$, we require $\int_N^{\infty} 1/x^4\,dx < 1/100$.
- **Test the Values from the Table:**
- • For N = 2: $1/24 \approx 0.0417 > 0.01$ (not guaranteed).
- • For N = 3: $1/81 \approx 0.0123 > 0.01$ (not guaranteed).
- • For N = 4: $1/192 \approx 0.0052 < 0.01$ (guaranteed!).
- **Conclusion:**
- • The smallest integer N that guarantees $R_N < 0.01$ is **N = 4**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Selects N = 3 because 1/81 is close to 0.01, but 0.0123 > 0.01, so it fails the strict bound.

✗ **(C)** N = 5 also satisfies the bound, but it is not the smallest integer.

✗ **(D)** Selects N = 2, which gives an error bound of over 4%.

Question 18 Analysis

★★★★ MANDATORY

Correct Answer: (B) It behaves like a constant multiple of the divergent harmonic series $\sum 1/n$.

STEP-BY-STEP SOLUTION

- **Compare with the Harmonic Series:**
- • The harmonic series is $\sum_{n=1}^{\infty} 1/n$, which diverges (p-series with $p = 1$).
- • Using the Limit Comparison Test with $b_n = 1/n$:
- • $\lim_{n\to\infty} [1/(3n+2)] / [1/n] = \lim_{n\to\infty} n/(3n+2) = 1/3$.
- • Because $0 < 1/3 < \infty$, both series share the same convergence behavior.
- • Since $\sum 1/n$ diverges, $\sum 1/(3n+2)$ **diverges**.
- • Alternatively, the Integral Test gives $\int_1^{\infty} 1/(3x+2)\,dx = [(1/3)\ln(3x+2)]_1^{\infty} = \infty$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** False: $\lim_{n\to\infty} 1/(3n+2) = 0$.

✗ **(C)** The series is not geometric; consecutive terms do not have a constant ratio.

✗ **(D)** All terms in the series are positive, so it is not an alternating series.

Question 19 Analysis

★★★ IMPORTANT

Correct Answer: (C) p > 2 and lim_(n→∞) 1 / n^(p/2) = 0

STEP-BY-STEP SOLUTION

- **Condition for p-Series Convergence:**
- • The series ∑ 1 / n^k converges if and only if exponent k > 1.
- • Here, exponent is k = p / 2 ⟹ p / 2 > 1 ⟹ p > 2.
- **Limit of Terms for Any Convergent Series:**
- • By the nth-Term Theorem, if any infinite series ∑ a_n converges, its terms must satisfy lim_(n→∞) a_n = 0.
- • For p > 2, lim_(n→∞) 1 / n^(p/2) = 0.
- • Thus, both **p > 2** and **lim a_n = 0** must hold.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Includes p = 2 (where p/2 = 1, giving the divergent harmonic series) and falsely claims lim a_n ≠ 0.
✗ **(B)** Fails to account for the division by 2, claiming p > 1 instead of p > 2.
✗ **(D)** Identifies the interval where the series diverges.

Question 20 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) I ≤ S ≤ f(1) + I

STEP-BY-STEP SOLUTION

- **Geometric Interpretation via Riemann Sums:**
- • Because f(x) is decreasing, consider rectangles of width 1:
- • **Left Riemann Sum:** f(1) + f(2) + f(3) + ... = S.
- The left rectangles lie above the curve f(x), so: $\int_1^\infty f(x)\,dx \le S$.
- Therefore, **I ≤ S**.
- • **Right Riemann Sum:** f(2) + f(3) + f(4) + ... = S – f(1).
- The right rectangles lie below the curve f(x), so: $S - f(1) \le \int_1^\infty f(x)\,dx = I$.
- Adding f(1) to both sides gives: **S ≤ f(1) + I**.
- **Combine the Two Inequalities:**
- • I ≤ S ≤ f(1) + I.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Arbitrarily divides by 2 without geometric foundation.
✗ **(B)** Claims the series sum is strictly greater than f(1) + I.
✗ **(C)** Falsely assumes the infinite sum S is smaller than the improper integral I.

Question 21 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) Direct Comparison Test with the convergent p-series ∑ 1 / n³

STEP-BY-STEP SOLUTION

- **Identify the Inequality for Direct Comparison:**
- • For all n ≥ 1: $n^3 + 5n + 2 > n^3 > 0$.
- • Taking the reciprocal reverses the inequality:
- • $0 < 1/(n^3 + 5n + 2) < 1/n^3$.
- **Evaluate the Benchmark Series:**
- • The benchmark series $\sum_{n=1}^{\infty} 1/n^3$ is a p-series with p = 3 > 1, which converges.
- **Apply the Direct Comparison Test (DCT):**
- • Because the terms $1/(n^3 + 5n + 2)$ are positive and strictly smaller than the terms of a known convergent p-series, $\sum_{n=1}^{\infty} 1/(n^3 + 5n + 2)$ **converges** by DCT.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Comparing a series with smaller terms to a divergent series yields no conclusion in DCT.
✗ **(C)** The nth-Term test can never be used to show convergence (it is inconclusive when lim a_n = 0).
✗ **(D)** The series is algebraic/polynomial, not geometric.

Question 22 Analysis

★★★★ MANDATORY

Correct Answer: (B) Diverges by the Direct Comparison Test because $1/(\sqrt{n}-0.5) > 1/\sqrt{n}$ and $\sum 1/\sqrt{n}$ diverges.

STEP-BY-STEP SOLUTION

- **Establish the Inequality for Direct Comparison:**
- • For all $n \geq 1$: $\sqrt{n} - 0.5 < \sqrt{n}$.
- • Taking reciprocals for positive terms:
- • $1/(\sqrt{n}-0.5) > 1/\sqrt{n} = 1/n^{(1/2)} > 0$.
- **Analyze the Benchmark Series:**
- • The series $\sum_{n=1}^{\infty} 1/n^{(1/2)}$ is a p-series with $p = 1/2 \leq 1$, which diverges.
- **Apply the Direct Comparison Test:**
- • Because the given terms are strictly greater than the terms of a divergent series with positive terms, $\sum_{n=1}^{\infty} 1/(\sqrt{n}-0.5)$ **diverges** by the Direct Comparison Test.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Reverses the inequality and incorrectly claims the p-series with $p = 1/2$ converges.
✗ **(C)** A p-series with $p = 1/2$ diverges, not converges, and the given series is not a pure p-series.
✗ **(D)** The limit of terms as $n \to \infty$ is 0, not ∞.

Question 23 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) Converges by the Limit Comparison Test with the convergent p-series $\sum 1/n^{(3/2)}$.

STEP-BY-STEP SOLUTION

- **Determine Dominant Behavior for Large n:**
- • Numerator dominant term: $2n^2$.
- • Denominator dominant term: $\sqrt{(n^7)} = n^{(7/2)}$.
- • As $n \to \infty$: $a_n \approx 2n^2 / n^{(7/2)} = 2/n^{(7/2-2)} = 2/n^{(3/2)}$.
- **Choose Benchmark Series:**
- • Let $b_n = 1/n^{(3/2)}$. This is a p-series with $p = 3/2 > 1$, which converges.
- **Compute the Limit Comparison:**
- • $L = \lim_{n\to\infty} a_n / b_n = \lim_{n\to\infty} [(2n^2+3)/\sqrt{(n^7+4n)}] \cdot n^{(3/2)}$
- • $= \lim_{n\to\infty} [2n^{(7/2)} + 3n^{(3/2)}] / \sqrt{(n^7+4n)} = 2/1 = 2$.
- **Apply the Limit Comparison Test:**
- • Since $0 < L = 2 < \infty$, both series share the same convergence behavior.
- • Because $\sum 1/n^{(3/2)}$ converges, the original series **converges**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** The Ratio Test on rational/algebraic functions always yields $L = 1$ (inconclusive)

Question 24 Analysis

★★★★ MANDATORY

Correct Answer: (D) Converges by the Direct Comparison Test because $0 \leq (3+\sin n)/n^2 \leq 4/n^2$.

STEP-BY-STEP SOLUTION

- **Bound the Numerator:**
- • Since $-1 \leq \sin n \leq 1$ for all real numbers n:
- • $3+(-1) \leq 3+\sin n \leq 3+1 \Rightarrow 2 \leq 3+\sin n \leq 4$.
- **Form the Comparison Inequality:**
- • Because $3+\sin n \geq 2 > 0$, all terms are strictly positive:
- • $0 < (3+\sin n)/n^2 \leq 4/n^2$ for all $n \geq 1$.
- **Analyze the Benchmark Series:**
- • $\sum_{n=1}^{\infty} 4/n^2 = 4\sum_{n=1}^{\infty} 1/n^2$ is a convergent p-series ($p = 2 > 1$).
- **Conclusion:**
- • By the Direct Comparison Test, $\sum_{n=1}^{\infty} (3+\sin n)/n^2$ **converges**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** $3+\sin n$ is always between 2 and 4, so terms are strictly positive.
✗ **(B)** Oscillation does not prevent convergence when bounded by a convergent p-series.
✗ **(C)** The denominator n^2 dominates n, making it smaller than the harmonic series, not larger.

Question 25 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $\lim_{n\to\infty} a_n / b_n = 0$ and $\sum b_n$ converges.

STEP-BY-STEP SOLUTION

- **Review the Standard Limit Comparison Test ($0 < L < \infty$):**
- • If $\lim (a_n / b_n) = L$ where $0 < L < \infty$, then both series either both converge or both diverge.
- **Review the Zero Limit Case ($L = 0$):**
- • If $\lim (a_n / b_n) = 0$, then a_n grows strictly slower than b_n ($a_n < b_n$ for sufficiently large n).
- • Therefore, if the larger benchmark series $\sum b_n$ **converges**, then $\sum a_n$ must also **converge**.
- • If $\sum b_n$ diverges when $L = 0$, the test is inconclusive.
- **Review the Infinite Limit Case ($L = \infty$):**
- • If $\lim (a_n / b_n) = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** When $L = 0$ and $\sum b_n$ diverges, the test is inconclusive.
✗ **(C)** When $L = \infty$ and $\sum b_n$ converges, the test is inconclusive.
✗ **(D)** If $L = 1$ and $\sum b_n$ diverges, then $\sum a_n$ diverges, not converges.

Question 26 Analysis

★★★★ MANDATORY

Correct Answer: (B) $b_n = 1 / n^2$; the series converges because $\lim_{n\to\infty} a_n / b_n = 1$ and $\sum 1 / n^2$ converges.

STEP-BY-STEP SOLUTION

- **Identify Leading Order of Magnitude:**
- • As $n \to \infty$: $(n + 1) / (n^3 - 4) \approx n / n^3 = 1 / n^2$.
- **Set Benchmark Series:**
- • $b_n = 1 / n^2$. This is a p-series with $p = 2 > 1$, so it converges.
- **Evaluate the Limit:**
- • $\lim_{n\to\infty} [(n + 1) / (n^3 - 4)] / [1 / n^2] = \lim_{n\to\infty} (n^3 + n^2) / (n^3 - 4) = 1$.
- **Conclusion via LCT:**
- • Because $L = 1$ (a finite positive number) and $\sum 1/n^2$ converges, the original series **converges**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Incorrectly simplifies the leading ratio as $n / n^2 = 1 / n$.
✗ **(C)** Using $b_n = 1/n^3$ gives limit ∞, not 0.
✗ **(D)** Contradicts the p-series result ($p = 2$ converges, not diverges).

Question 27 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) The inequality $1 / (n^2 - 0.5) > 1 / n^2$ holds, so the terms are larger than the convergent series, making DCT inconclusive in that direction.

STEP-BY-STEP SOLUTION

- **Evaluate the Inequality Between the Series Terms:**
- • For all $n \geq 1$: $n^2 - 0.5 < n^2$.
- • Taking reciprocals reverses the inequality:
- • $1 / (n^2 - 0.5) > 1 / n^2$.
- **Recall the Rule of the Direct Comparison Test:**
- • To prove convergence via DCT, the terms of the target series must be *less than or equal to* the terms of a known convergent series: $0 \leq a_n \leq b_n$.
- • Here, $a_n > b_n$, which gives no information (being larger than a finite sum does not prove finiteness).
- • To properly prove convergence, one must use the Limit Comparison Test with $1 / n^2$ or DCT with $2 / n^2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** For $n \geq 1$, $n^2 \geq 1$, so $n^2 - 0.5 \geq 0.5 > 0$ (all terms are positive).
✗ **(B)** False: $\sum 1/n^2$ converges ($p = 2 > 1$).
✗ **(D)** Direct comparison can be used if an appropriate smaller upper bound is found.

Question 28 Analysis

★★★★ MANDATORY

Correct Answer: (D) The series ∑ a_n converges, and its sum satisfies 0 < S < 1.

STEP-BY-STEP SOLUTION

- **Verify the DCT Conditions:**
- • From the table and algebraic definition: $2^n + 1 > 2^n > 0 \Rightarrow 0 < a_n < b_n$ for all $n \geq 1$.
- **Evaluate the Benchmark Sum:**
- • $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} (1/2)^n = (1/2) / [1 - 1/2] = 1$.
- **Apply the Direct Comparison Test:**
- • Because $0 < a_n < b_n$ for all n, the series $\sum a_n$ converges.
- • Furthermore, term-by-term strict inequality ensures that the infinite sum satisfies: $0 < S = \sum a_n < \sum b_n = 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** DCT does not require a constant ratio; that is only for the Geometric Series Test.
✗ **(B)** Contradicts DCT: terms bounded by a convergent geometric series cannot grow without bound.
✗ **(C)** The sum cannot equal 1 because every single term a_n is strictly smaller than b_n.

Question 29 Analysis

★★★★ MANDATORY

Correct Answer: (A) Converges by the Limit Comparison Test with ∑ 1 / n^(3/2)

STEP-BY-STEP SOLUTION

- **Find the Asymptotic Behavior of the Terms:**
- • As $n \to \infty$: $\sqrt{(n^3 + 2n)} \approx \sqrt{(n^3)} = n^{(3/2)}$.
- • Therefore, a_n behaves like $b_n = 1 / n^{(3/2)}$.
- **Apply Limit Comparison Test:**
- • $\lim_{n\to\infty} [1 / \sqrt{(n^3 + 2n)}] / [1 / n^{(3/2)}] = \lim_{n\to\infty} \sqrt{(n^3 / (n^3 + 2n))} = \sqrt{(1)} = 1$.
- • Since $L = 1$ is finite and positive, both series share the same convergence behavior.
- **Analyze Benchmark Series:**
- • $\sum 1 / n^{(3/2)}$ is a p-series with $p = 3/2 > 1$, so it converges.
- • Therefore, $\sum_{n=1}^{\infty} 1 / \sqrt{(n^3 + 2n)}$ **converges**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Fails to take the square root of n^3, incorrectly guessing the benchmark is $1/n^3$.
✗ **(C)** A p-series with $p = 3/2$ converges ($p > 1$), not diverges.
✗ **(D)** The Ratio Test yields $L = 1$ on algebraic functions, which is inconclusive.

Question 30 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) Direct Comparison Test: $0 < \arctan(n) / n^3 < (\pi / 2) / n^3$, and $(\pi / 2) \sum 1 / n^3$ converges.

STEP-BY-STEP SOLUTION

- **Bound the Arc Tangent Function:**
- • For all $n \geq 1$: $0 < \arctan(1) \leq \arctan(n) < \pi / 2$.
- **Set Up the DCT Upper Bound:**
- • $0 < \arctan(n) / n^3 < (\pi / 2) / n^3$ for all $n \geq 1$.
- **Evaluate the Benchmark Series:**
- • The benchmark series is $(\pi / 2) \sum_{n=1}^{\infty} 1 / n^3$.
- • This is a constant multiple of a convergent p-series with $p = 3 > 1$, so it converges.
- **Conclusion:**
- • By the Direct Comparison Test, $\sum_{n=1}^{\infty} \arctan(n) / n^3$ **converges**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** The nth-Term test can never be used to conclude convergence.
✗ **(C)** The harmonic series ∑ 1/n diverges, not converges.
✗ **(D)** The series is not geometric.

Question 31 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) It converges by the Alternating Series Test because $1/\sqrt{n} > 0$, $1/\sqrt{(n+1)} \le 1/\sqrt{n}$, and $\lim_{n\to\infty} 1/\sqrt{n} = 0$.

STEP-BY-STEP SOLUTION

- **Identify the Magnitude Sequence b_n:**
- • The series is alternating: $\sum (-1)^{n+1} b_n$ with $b_n = 1/\sqrt{n}$.
- **Verify the Three Conditions of the Alternating Series Test (AST):**
- • 1. **Positive Terms:** $b_n = 1/\sqrt{n} > 0$ for all $n \ge 1$.
- • 2. **Decreasing Terms:** Since $n+1 > n \Rightarrow \sqrt{(n+1)} > \sqrt{n} \Rightarrow 1/\sqrt{(n+1)} < 1/\sqrt{n}$, so $b_{(n+1)} \le b_n$.
- • 3. **Limit Equals Zero:** $\lim_{n\to\infty} 1/\sqrt{n} = 0$.
- **Conclusion:**
- • All three hypotheses are strictly satisfied, so the series **converges** by the Alternating Series Test.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** The absolute series $\sum 1/\sqrt{n}$ diverges ($p = 1/2 \le 1$), so it does not converge absolutely.
✗ **(B)** A p-series with $p = 1/2$ diverges; the series only converges due to alternation.
✗ **(D)** The Ratio Test on algebraic powers yields $L = 1$, which is inconclusive.

Question 32 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) 1 / 36

STEP-BY-STEP SOLUTION

- **State the Alternating Series Estimation Theorem:**
- • If an alternating series $\sum (-1)^{n+1} b_n$ satisfies the conditions of the Alternating Series Test, the error in approximating the infinite sum S by the N-th partial sum S_N is bounded by the magnitude of the first omitted term:
- • $|R_N| = |S - S_N| \le b_{(N+1)}$.
- **Evaluate for N = 5:**
- • The first omitted term corresponds to $n = 6$:
- • $b_6 = 1/6^2 = 1/36$.
- **Conclusion:**
- • The maximum possible error is **1 / 36**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to square the denominator, evaluating 1 / 6 instead of 1 / 36.
✗ **(B)** Uses the last included term $b_5 = 1/25$ instead of the first omitted term.
✗ **(C)** Uses the second omitted term $b_7 = 1/49$.

Question 33 Analysis

★★★★ MANDATORY

Correct Answer: (A) An underestimate, because the first omitted term is positive (+1 / 11).

STEP-BY-STEP SOLUTION

- **Inspect the Sequence of Partial Sums:**
- • For an alternating series starting with a positive term whose magnitudes decrease to 0:
- • Odd partial sums ($S_1, S_3, S_5, \ldots$) are overestimates ($S_{odd} > S$).
- • Even partial sums ($S_2, S_4, S_6, \ldots$) are underestimates ($S_{even} < S$).
- **Analyze the First Omitted Term:**
- • S_4 ends with the negative term $-1/9$ ($n = 4$).
- • The next term added to S_4 to get S_5 is positive: $a_5 = (-1)^6/(2(5)+1) = +1/11 > 0$.
- • Because $S = S_4 + 1/11 - 1/13 + \ldots$, adding a positive quantity that exceeds the subsequent negative ones guarantees that $S > S_4$.
- • Therefore, S_4 is an **underestimate** of the true sum S.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Ending on a negative term means the approximation was pulled below the true sum, making it an underestimate, not an overestimate.
✗ **(C)** Partial sums alternate between over- and underestimates; they are never all overestimates.
✗ **(D)** A finite partial sum of an infinite series is an approximation, not the exact sum.

Question 34 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 10

STEP-BY-STEP SOLUTION

- **Set Up the Error Bound Inequality:**
- • By the Alternating Series Error Bound: $|R_N| \le b_{N+1} = 1/(N+1)^3$.
- • We require: $1/(N+1)^3 < 0.001 = 1/1000$.
- **Solve for N:**
- • Taking reciprocals: $(N+1)^3 > 1000$.
- • Take the cube root of both sides:
- • $N+1 > \sqrt[3]{1000} = 10$.
- • $N > 9 \Longrightarrow$ Since N must be an integer, $N \ge 10$.
- **Conclusion:**
- • The smallest positive integer is **N = 10** (which gives $b_{11} = 1/1331 < 1/1000$).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Chooses N = 9, but for N = 9, the error bound is $1/(9+1)^3 = 1/1000$, which is NOT strictly less than 0.001.

✗ **(C)** N = 11 works, but it is not the smallest integer.

✗ **(D)** Uses a linear approximation instead of cubing N + 1.

Question 35 Analysis

★★★★ MANDATORY

Correct Answer: (C) Alternating Series Test, because $\cos(\pi n) = (-1)^n$ and $1/n$ decreases monotonically to 0.

STEP-BY-STEP SOLUTION

- **Simplify the Trigonometric Factor:**
- • For integer values of n:
- • When n = 1: $\cos(\pi) = -1$.
- • When n = 2: $\cos(2\pi) = 1$.
- • When n = 3: $\cos(3\pi) = -1$.
- • In general: $\cos(\pi n) = (-1)^n$.
- **Rewrite the Series:**
- • $\sum_{n=1}^{\infty} \cos(\pi n)/n = \sum_{n=1}^{\infty} (-1)^n/n$.
- • This is the alternating harmonic series.
- **Apply the Alternating Series Test:**
- • $b_n = 1/n > 0$, $b_{n+1} \le b_n$, and $\lim_{n\to\infty} 1/n = 0$.
- • Therefore, the series converges by the **Alternating Series Test**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** The terms $(-1)^n/n$ approach 0 as $n \to \infty$, so the Divergence Test does not apply.

Question 36 Analysis

★★★★ MANDATORY

Correct Answer: (D) It diverges by the nth-Term Test because $\lim_{n\to\infty} a_n$ does not exist (the terms oscillate near ±3/2).

STEP-BY-STEP SOLUTION

- **Examine the Limit of the Sequence of Terms:**
- • Consider the absolute magnitude of the terms: $b_n = (3n+1)/(2n-1)$.
- • $\lim_{n\to\infty} b_n = \lim_{n\to\infty} (3+1/n)/(2-1/n) = 3/2$.
- • Because $b_n \to 3/2 \neq 0$, the terms of the series $a_n = (-1)^n b_n$ oscillate between values approaching +3/2 and −3/2.
- **Apply the nth-Term Test for Divergence:**
- • $\lim_{n\to\infty} a_n$ does not exist (and specifically is not 0).
- • By the nth-Term Test for Divergence, the series must **diverge**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** The series does not converge because the terms fail to approach zero.

✗ **(B)** The Alternating Series Test can only show convergence; it cannot be cited to prove divergence.

✗ **(C)** This is a rational sequence, not a p-series.

Question 37 Analysis

★★★★ MANDATORY

Correct Answer: (A) [0.583, 0.783]

STEP-BY-STEP SOLUTION

- **Recall Alternating Series Partial Sum Properties:**
- • $S_4 = 0.583$ is an even partial sum, which is an underestimate: $S_4 \le S$.
- • The error in S_4 is bounded by the next term: $|S - S_4| \le b_5 = 0.200$.
- **Set Up the Inequality for S:**
- • $S_4 \le S \le S_4 + b_5$.
- • $0.583 \le S \le 0.583 + 0.200 = 0.783$.
- **Conclusion:**
- • The exact sum S is guaranteed to lie in the interval **[0.583, 0.783]**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Subtracts b_5 from S_4, which corresponds to the incorrect direction for an even partial sum.
✗ **(C)** Uses the first two terms $[S_2, S_1]$, which is unnecessarily broad and does not utilize S_4 and b_5.
✗ **(D)** Arbitrarily combines upper limits without adhering to the error bound.

Question 38 Analysis

★★★★ MANDATORY

Correct Answer: (B) 1 / 720

STEP-BY-STEP SOLUTION

- **Identify the First Omitted Term:**
- • The approximation uses terms up to n = 2.
- • The first omitted term corresponds to n = 3:
- • $a_3 = (-1)^4 / [2(3)]! = 1/6!$.
- **Compute the Value of 6!:**
- • $6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$.
- • Thus, $b_3 = 1/720$.
- **Apply the Alternating Series Error Bound:**
- • $|S - S_2| \le b_3 =$ **1 / 720**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Computes $5! = 120$ instead of $(2 \cdot 3)! = 6! = 720$.
✗ **(C)** Reports the magnitude of the last included term $b_2 = 1/4! = 1/24$.
✗ **(D)** Computes $7! = 5040$.

Question 39 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) Without monotonicity, the positive terms can sum to infinity while the negative terms sum to a finite value, causing the series to diverge.

STEP-BY-STEP SOLUTION

- **Understand the Necessity of the Decreasing Condition:**
- • Consider the famous counterexample series:
- • $a_n = 1/\sqrt{2} - 1/2 + 1/\sqrt{3} - 1/3 + 1/\sqrt{4} - 1/4 + \ldots$
- • Here, the signs alternate and $\lim a_n = 0$.
- • However, the positive terms $\sum 1/\sqrt{n}$ diverge much faster than the negative terms $\sum 1/n$, so the overall sum diverges to $+\infty$!
- • The condition **$b_{n+1} \le b_n$** ensures that each positive term is paired with a subsequent term that offsets it, guaranteeing that the sequence of partial sums converges.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Absurd claim confusing monotonicity with geometric ratios.
✗ **(B)** False: as demonstrated by the counterexample, alternating signs and $\lim a_n = 0$ alone do not guarantee convergence.
✗ **(D)** The condition guarantees convergence, not the sign of the sum.

Question 40 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) 1 / 384

STEP-BY-STEP SOLUTION

- **Identify the First Omitted Term:**
- • The approximation uses the first two terms: 1 and $-(0.5)^2 / 2!$.
- • The first omitted term is the third term of the series:
- • $b_3 = (0.5)^4 / 4!$.
- **Evaluate the Third Term:**
- • $0.5 = 1/2 \Longrightarrow (0.5)^4 = (1/2)^4 = 1/16$.
- • $4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$.
- • $b_3 = (1/16) / 24 = 1/(16 \cdot 24) = 1/384$.
- **Apply the Alternating Series Error Bound:**
- • $|\cos(0.5) - 0.875| \leq b_3 =$ **1 / 384**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Divides by an extra power of 2.
✗ **(B)** Multiplies by 12 instead of 24 (forgetting a factor of 2 in 4!).
✗ **(C)** Uses $2! \cdot 24$ or miscalculates $(0.5)^4$.

Question 41 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) Conditionally convergent

STEP-BY-STEP SOLUTION

- **Test the Alternating Series for Convergence:**
- • The series is the alternating harmonic series: $1 - 1/2 + 1/3 - 1/4 + \ldots$.
- • $b_n = 1/n$ is positive, decreasing $(1/(n+1) < 1/n)$, and $\lim_{n\to\infty} 1/n = 0$.
- • By the Alternating Series Test, the series converges.
- **Test the Series of Absolute Values:**
- • Consider $\sum_{n=1}^{\infty} |(-1)^{n+1} / n| = \sum_{n=1}^{\infty} 1/n$.
- • This is the harmonic series (p-series with $p = 1$), which diverges.
- **Conclusion:**
- • Because the series converges, but does not converge absolutely, it is **conditionally convergent**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Fails to recognize that the absolute series $\sum 1/n$ diverges.
✗ **(C)** Ignores the Alternating Series Test which guarantees convergence.
✗ **(D)** The AST is completely conclusive.

Question 42 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) Absolutely convergent because $\sum 1/n^3$ converges by the p-series test.

STEP-BY-STEP SOLUTION

- **Evaluate the Series of Absolute Values:**
- • $\sum_{n=1}^{\infty} |(-1)^n / n^3| = \sum_{n=1}^{\infty} 1/n^3$.
- • This is a p-series with $p = 3 > 1$, which converges.
- **Definition of Absolute Convergence:**
- • If the series of absolute values $\sum |a_n|$ converges, then the original series $\sum a_n$ is defined to be **absolutely convergent**.
- • By theorem, absolute convergence implies convergence.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Alternating signs do not make a series conditionally convergent if its absolute values also converge.
✗ **(C)** The terms approach 0 as $n \to \infty$, so it does not diverge by the nth-Term test.
✗ **(D)** $p = 3 > 1$ proves absolute convergence, not conditional convergence.

Question 43 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) Converges absolutely because L = lim_(n→∞) |a_(n+1) / a_n| = 0 < 1.

STEP-BY-STEP SOLUTION

- **Set Up the Ratio Test:**
- • L = lim_(n→∞) | a_(n+1) / a_n |.
- • a_(n+1) = 3^{n+1} / (n + 1)! and a_n = 3^n / n!.
- **Simplify the Ratio:**
- • | a_(n+1) / a_n | = [3^{n+1} / (n + 1)!] · [n! / 3^n]
- • = [3^{n+1} / 3^n] · [n! / (n + 1)!] = 3 · [1 / (n + 1)] = 3 / (n + 1).
- **Evaluate the Limit as n → ∞:**
- • L = lim_(n→∞) 3 / (n + 1) = 0.
- **Conclusion:**
- • Because L = 0 < 1, the series **converges absolutely** by the Ratio Test.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Ratio Test yields L = 0, which is conclusive and proves convergence.
✗ **(B)** Forgets to simplify (n+1)!, leaving 3 as the limit.
✗ **(D)** Factorial growth n! dominates exponential growth 3^n for large n.

Question 44 Analysis

★★★★ MANDATORY

Correct Answer: (D) The series diverges because L = 5 > 1.

STEP-BY-STEP SOLUTION

- **Set Up the Ratio:**
- • | a_(n+1) / a_n | = [5^{n+1} / $(n + 1)^3$] · [n^3 / 5^n].
- **Group Like Terms:**
- • = [5^{n+1} / 5^n] · [n / (n + 1)]^3 = 5 · [n / (n + 1)]^3.
- **Evaluate the Limit:**
- • L = lim_(n→∞) 5 · [n / (n + 1)]^3 = 5 · $(1)^3$ = 5.
- **Conclusion:**
- • Because L = 5 > 1, the series **diverges** by the Ratio Test.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** The series is not a p-series because of the exponential numerator 5^n.
✗ **(B)** Inverts the ratio, obtaining 1/5 instead of 5.
✗ **(C)** Ignores the exponential factor 5^n, looking only at n^3.

Question 45 Analysis

★★★★ MANDATORY

Correct Answer: (A) $\sum_{n=1}^{\infty}$ (n^2+ 1) / (n^4+ 3)

STEP-BY-STEP SOLUTION

- **Recall When the Ratio Test Fails (L = 1):**
- • The Ratio Test always yields L = 1 for any series whose terms are rational or algebraic functions of n (polynomials, roots, p-series).
- **Test Option A:**
- • a_n = (n^2 + 1) / (n^4 + 3).
- • lim |a_(n+1) / a_n| = lim [((n+1)^2 + 1) / ((n+1)^4 + 3)] · [(n^4 + 3) / (n^2 + 1)] = 1.
- • Thus, the Ratio Test is **inconclusive** on Option A (convergence must be determined by LCT with 1/n^2).
- **Verify Other Options:**
- • Option B: L = 0 (converges).
- • Option C: L = 3/4 (converges).
- • Option D: L = ∞ (diverges).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Factorial series yields L = 0 < 1 (conclusive).
✗ **(C)** Geometric series yields L = 3/4 < 1 (conclusive).
✗ **(D)** Factorial numerator yields L = ∞ > 1 (conclusive).

Question 46 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) Converges conditionally

STEP-BY-STEP SOLUTION

- **Test Alternating Series Convergence:**
- • $b_n = 1/\sqrt{(n^2+1)}$ is positive, strictly decreasing, and $\lim_{n\to\infty} 1/\sqrt{(n^2+1)} = 0$.
- • By the Alternating Series Test, the series converges.
- **Test Absolute Convergence:**
- • Consider $\sum |a_n| = \sum 1/\sqrt{(n^2+1)}$.
- • For large n: $1/\sqrt{(n^2+1)}$ behaves like $1/n$.
- • Using Limit Comparison Test with the divergent harmonic series $b_n = 1/n$:
- • $\lim_{n\to\infty} [1/\sqrt{(n^2+1)}]/[1/n] = \lim_{n\to\infty} n/\sqrt{(n^2+1)} = 1$.
- • Since $L = 1$ and $\sum 1/n$ diverges, the absolute series $\sum 1/\sqrt{(n^2+1)}$ diverges.
- **Conclusion:**
- • The series converges, but not absolutely; therefore, it is **conditionally convergent**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Mistakenly thinks $1/\sqrt{(n^2+1)}$ behaves like $1/n^2$ ($p = 2$).

✗ **(C)** The terms approach 0, so the nth-Term test does not show divergence.

Question 47 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) Converges absolutely because $L = 1/4 < 1$

STEP-BY-STEP SOLUTION

- **Set Up the Ratio $|a_{n+1}/a_n|$:**
- • $a_{n+1} = [(n+1)!]^2/[2(n+1)]! = [(n+1)!]^2/(2n+2)!$.
- • $a_n = (n!)^2/(2n)!$.
- **Divide and Group Like Terms:**
- • $|a_{n+1}/a_n| = [((n+1)!)^2/(n!)^2] \cdot [(2n)!/(2n+2)!]$.
- • Notice: $(n+1)!/n! = n+1 \Longrightarrow [(n+1)!/n!]^2 = (n+1)^2$.
- • Notice: $(2n+2)! = (2n+2)(2n+1)(2n)!$, so $(2n)!/(2n+2)! = 1/[(2n+2)(2n+1)]$.
- **Compute the Limit as $n \to \infty$:**
- • $L = \lim_{n\to\infty} (n+1)^2/[(2n+2)(2n+1)] = \lim_{n\to\infty} (n^2+2n+1)/(4n^2+6n+2) = 1/4$.
- **Conclusion:**
- • Since $L = 1/4 < 1$, the series **converges absolutely**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to square (n+1) or divides by 2 instead of 4.

✗ **(B)** Inverts the ratio, obtaining $L = 4$.

Question 48 Analysis

★★★★ MANDATORY

Correct Answer: (D) The series converges absolutely because $\lim_{n\to\infty} |a_{n+1}/a_n| = 0.700 < 1$.

STEP-BY-STEP SOLUTION

- **Identify the Limit from the Table:**
- • As n increases from 1 to 100, the ratio $|a_{n+1}/a_n|$ approaches $L = 0.700$.
- **Apply the Ratio Test Criteria:**
- • 1. If $L < 1$, the series converges absolutely.
- • 2. If $L > 1$, the series diverges.
- • 3. If $L = 1$, the test is inconclusive.
- **Form the Conclusion:**
- • Here, $L = 0.700 < 1$. Therefore, the series **converges absolutely**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** L is the ratio limit, not the sum of the series, and the convergence is absolute, not conditional.

✗ **(B)** The terms of a convergent series can have non-zero ratios as long as $L < 1$.

✗ **(C)** The test is only inconclusive when $L = 1$.

Question 49 Analysis

★★★★ MANDATORY

Correct Answer: (A) The series $\sum_{n=1}^{\infty} a_n$ converges, and $|\sum_{n=1}^{\infty} a_n| \le \sum_{n=1}^{\infty} |a_n|$.

STEP-BY-STEP SOLUTION

- **Theorem: Absolute Convergence Implies Convergence:**
- • If $\sum |a_n|$ converges, then $\sum a_n$ also converges.
- **Triangle Inequality for Infinite Series:**
- • By the generalized triangle inequality:
- • $|a_1 + a_2 + a_3 + \ldots| \le |a_1| + |a_2| + |a_3| + \ldots$
- • Therefore: $|\sum_{n=1}^{\infty} a_n| \le \sum_{n=1}^{\infty} |a_n|$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** A series cannot be both absolutely convergent and conditionally convergent; they are mutually exclusive.

✗ **(C)** Absolute convergence guarantees convergence, never divergence.

✗ **(D)** For any convergent series, $\lim_{n\to\infty} a_n = 0$, never 1.

Question 50 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) (II) only

STEP-BY-STEP SOLUTION

- **Analyze Series (I):**
- • $\sum |(-1)^n / n^2| = \sum 1/n^2$ ($p = 2 > 1$ converges).
- • Thus, (I) is **absolutely convergent**.
- **Analyze Series (II):**
- • Converges by AST (alternating harmonic series).
- • $\sum |(-1)^n / n| = \sum 1/n$ (harmonic series, $p = 1$ diverges).
- • Thus, (II) is **conditionally convergent**.
- **Analyze Series (III):**
- • $\lim_{n\to\infty} |a_n| = \lim_{n\to\infty} n/(n+1) = 1 \neq 0$.
- • By the nth-Term Test for Divergence, (III) **diverges**.
- **Conclusion:**
- • Only **(II)** converges conditionally.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Includes (I), which is absolutely convergent, not conditionally convergent.

Question 51 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) $(x-1) - (1/2)(x-1)^2 + (1/3)(x-1)^3$

STEP-BY-STEP SOLUTION

- **Compute the Function and Derivatives at c = 1:**
- • $f(x) = \ln(x) \Rightarrow f(1) = \ln(1) = 0$.
- • $f'(x) = 1/x \Rightarrow f'(1) = 1$.
- • $f''(x) = -1/x^2 \Rightarrow f''(1) = -1$.
- • $f'''(x) = 2/x^3 \Rightarrow f'''(1) = 2$.
- **Apply the Taylor Polynomial Formula:**
- • $P_3(x) = f(1) + f'(1)(x-1) + [f''(1)/2!](x-1)^2 + [f'''(1)/3!](x-1)^3$.
- **Substitute the Values:**
- • $P_3(x) = 0 + 1(x-1) + [-1/2](x-1)^2 + [2/6](x-1)^3$
- • $= (x-1) - (1/2)(x-1)^2 + (1/3)(x-1)^3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign errors in the derivatives of 1/x.

✗ **(B)** Forgets to divide by the factorials 2! and 3!.

✗ **(D)** Incorrectly sets $f(1) = 1$ instead of $\ln(1) = 0$.

Question 52 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) −24

STEP-BY-STEP SOLUTION

- **Relate Polynomial Coefficients to Derivatives:**
- • In any Maclaurin polynomial $P_n(x) = \sum [f^{(k)}(0) / k!] x^k$, the coefficient of x^k is given by:
- • $a_k = f^{(k)}(0) / k!$.
- **Identify the Coefficient of x^3:**
- • For $k = 3$, the term in $P_4(x)$ is $-4x^3$.
- • Therefore: $a_3 = -4$.
- **Solve for $f'''(0)$:**
- • $f'''(0) / 3! = -4$.
- • $f'''(0) = -4 \cdot 3! = -4 \cdot (6) = \mathbf{-24}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Computes $f^{(4)}(0) = 7 \cdot 4! / 4$ or makes an arithmetic error.

✗ **(B)** Reports the coefficient −4 directly, forgetting to multiply by 3!.

✗ **(C)** Multiplies by 3 instead of $3! = 6$.

Question 53 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 1 / 32

STEP-BY-STEP SOLUTION

- **State the Lagrange Error Bound Formula:**
- • For an n-th degree Taylor polynomial centered at c:
- • $|R_n(x)| = |f(x) - P_n(x)| \le [M / (n+1)!] |x - c|^{(n+1)}$, where $M = \max |f^{(n+1)}(t)|$ between c and x.
- **Substitute Given Values (n = 3, c = 2, x = 2.5):**
- • $n + 1 = 4$.
- • $(n + 1)! = 4! = 24$.
- • $|x - c| = |2.5 - 2| = 0.5 = 1 / 2$.
- • $|x - c|^4 = (1 / 2)^4 = 1 / 16$.
- • $M = 12$.
- **Compute the Bound:**
- • $|R_3(2.5)| \le [12 / 24] \cdot (1 / 16) = (1 / 2) \cdot (1 / 16) = \mathbf{1 / 32}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses $3! = 6$ instead of $4! = 24$ in the denominator: $(12 / 6)(1/16) = 2/16 = 1/8$, or forgets factor of 1/2.

✗ **(C)** Divides by an extra power of 2.

Question 54 Analysis

★★★★ MANDATORY

Correct Answer: (B) 0.82

STEP-BY-STEP SOLUTION

- **Form the Second-Degree Maclaurin Polynomial for e^x:**
- • $P_2(x) = 1 + x + x^2 / 2! = 1 + x + x^2 / 2$.
- **Substitute x = −0.2:**
- • $P_2(-0.2) = 1 + (-0.2) + (-0.2)^2 / 2$.
- • $(-0.2)^2 = 0.04$.
- • $P_2(-0.2) = 1 - 0.2 + 0.04 / 2 = 0.8 + 0.02 = \mathbf{0.82}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits the quadratic term: $1 - 0.2 = 0.80$.

✗ **(C)** Forgets to divide by 2! in the quadratic term: $0.8 + 0.04 = 0.84$.

✗ **(D)** Subtracts the quadratic term: $0.8 - 0.02 = 0.78$.

Question 55 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 1 / 750

STEP-BY-STEP SOLUTION

- **Find the Maximum of the Third Derivative on [−0.2, 0]:**
- • $f'''(t) = d^3/dt^3 (e^t) = e^t$.
- • Because e^t is an increasing function, its maximum on [−0.2, 0] occurs at the right endpoint t = 0:
- • $M = \max |e^t| = e^0 = 1$.
- **Apply the Lagrange Error Bound Formula (n = 2):**
- • $|R_2(-0.2)| \le [M / 3!] |-0.2|^3$.
- • $3! = 6$.
- • $|-0.2|^3 = (2 / 10)^3 = (1 / 5)^3 = 1 / 125$.
- **Multiply:**
- • $|R_2| \le (1 / 6) \cdot (1 / 125) = 1 / 750$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses 1/50 instead of 1/125.
✗ **(B)** Uses 2! = 2 instead of 3! = 6 in the denominator: (1/2)(1/125) = 1/250, or arithmetic error.
✗ **(D)** Omits the 3! factor completely: 1 / 125.

Question 56 Analysis

★★★★ MANDATORY

Correct Answer: (D) $2 + (1 / 4)(x - 4) - (1 / 64)(x - 4)^2$

STEP-BY-STEP SOLUTION

- **Compute Derivatives at c = 4:**
- • $f(x) = x^{(1/2)} \Rightarrow f(4) = \sqrt{4} = 2$.
- • $f'(x) = (1/2) x^{(-1/2)} = 1 / (2\sqrt{x}) \Rightarrow f'(4) = 1 / (2 \cdot 2) = 1 / 4$.
- • $f''(x) = -(1/4) x^{(-3/2)} = -1 / (4 x^{(3/2)}) \Rightarrow f''(4) = -1 / (4 \cdot 8) = -1 / 32$.
- **Form the Taylor Polynomial:**
- • $P_2(x) = f(4) + f'(4)(x - 4) + [f''(4) / 2!] (x - 4)^2$.
- • $P_2(x) = 2 + (1 / 4)(x - 4) + [-1 / (32 \cdot 2)] (x - 4)^2$
- • $= 2 + (1 / 4)(x - 4) - (1 / 64)(x - 4)^2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error on the second derivative of x^(1/2).
✗ **(B)** Forgets to divide by 2! in the quadratic term, leaving −1/32.
✗ **(C)** Evaluates f'(4) = 1/2 instead of 1/4.

Question 57 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $5 - 3(x - 2) + 4(x - 2)^2 - 2(x - 2)^3$

STEP-BY-STEP SOLUTION

- **Formula for Third-Degree Taylor Polynomial Centered at c = 2:**
- • $P_3(x) = f(2) + f'(2)(x - 2) + [f''(2) / 2!](x - 2)^2 + [f'''(2) / 3!](x - 2)^3$.
- **Substitute the Given Values:**
- • $f(2) = 5$.
- • $f'(2) = -3 \Rightarrow$ term is $-3(x - 2)$.
- • $f''(2) / 2! = 8 / 2 = 4 \Rightarrow$ term is $+4(x - 2)^2$.
- • $f'''(2) / 3! = -12 / 6 = -2 \Rightarrow$ term is $-2(x - 2)^3$.
- **Combine Terms:**
- • $P_3(x) = 5 - 3(x - 2) + 4(x - 2)^2 - 2(x - 2)^3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Omits the factorial denominators 2! and 3!.
✗ **(C)** Centers the polynomial at 0 instead of c = 2.
✗ **(D)** Divides f'''(2) by 2! instead of 3! = 6.

Question 58 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 1 / 80

STEP-BY-STEP SOLUTION

- **Identify Parameters for Lagrange Error Bound:**
- • $n = 4$, $c = 0$, $x = 0.5 = 1/2$.
- • $M = \max |f^{(5)}(t)| = 48$.
- • $(n+1)! = 5! = 120$.
- • $|x - c|^{(n+1)} = (1/2)^5 = 1/32$.
- **Calculate the Bound:**
- • $|R_4(0.5)| \le [M/5!]\,|0.5|^5 = [48/120] \cdot (1/32)$.
- • Simplify $48/120 = 2/5$.
- • $|R_4(0.5)| \le (2/5) \cdot (1/32) = 1/(5 \cdot 16) = \mathbf{1/80} = 0.0125$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses $(0.5)^4$ instead of $(0.5)^5$.
✗ **(C)** Omits the numerator 48, using $1/120$.
✗ **(D)** Divides by an extra factor of 2.

Question 59 Analysis

★★★★ MANDATORY

Correct Answer: (C) f has a relative maximum at x = 3 because f'(3) = 0 and f''(3) = −8 < 0.

STEP-BY-STEP SOLUTION

- **Read the Values from the Taylor Polynomial:**
- • In $P_2(x) = f(3) + f'(3)(x-3) + [f''(3)/2!](x-3)^2$:
- • Constant term: $f(3) = 7$.
- • Linear term: coefficient of $(x-3)$ is $0 \Rightarrow f'(3) = 0$ (critical point).
- • Quadratic term: $f''(3)/2! = -4 \Rightarrow f''(3) = -8$.
- **Apply the Second Derivative Test:**
- • Since $f'(3) = 0$ and $f''(3) = -8 < 0$, the curve is concave down at the horizontal tangent.
- • Therefore, f has a **relative maximum** at $x = 3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** An inflection point requires $f''(3) = 0$ with a sign change, but $f''(3) = -8 \neq 0$.
✗ **(B)** Concave down ($f'' < 0$) corresponds to a relative maximum, not a relative minimum.
✗ **(D)** Since $f'(3) = 0$, f is not strictly increasing at $x = 3$.

Question 60 Analysis

★★★★ MANDATORY

Correct Answer: (D) The Alternating Series Error Bound only requires evaluating the magnitude of the first omitted term, without maximizing an unknown derivative.

STEP-BY-STEP SOLUTION

- **Compare the Two Error Estimation Techniques:**
- • The **Lagrange Error Bound** requires determining $M = \max |f^{(n+1)}(t)|$ over an interval, which often demands maximizing an unfamiliar derivative function.
- • In contrast, when a Taylor series is alternating and satisfies the Alternating Series Test, the **Alternating Series Error Bound** guarantees: $|R_n| \le b_{(n+1)}$.
- • This bound is calculated directly from the very next term in the polynomial without requiring any derivative maximization.
- • Thus, Option A accurately captures the primary computational advantage.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Both methods provide upper bounds on error, not the exact error value.
✗ **(B)** Depending on the function, one bound may be tighter than the other; neither is universally smaller.
✗ **(C)** The Lagrange Error Bound applies to all Taylor polynomials, whether alternating or not.

Question 61 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) [−1, 5)

STEP-BY-STEP SOLUTION

- **Apply the Ratio Test to Find the Radius of Convergence:**
- • $L = \lim_{n\to\infty} |a_{n+1} / a_n| = \lim_{n\to\infty} |[(x-2)^{n+1} / ((n+1)\cdot 3^{n+1})] \cdot [(n\cdot 3^n)/(x-2)^n]|$.
- • $= \lim_{n\to\infty} |x-2|/3 \cdot [n/(n+1)] = |x-2|/3$.
- • Set $L < 1$: $|x-2|/3 < 1 \Rightarrow |x-2| < 3$ (Radius $R = 3$).
- • $-3 < x-2 < 3 \Rightarrow -1 < x < 5$.
- **Test the Left Endpoint x = −1:**
- • Substitute $x = -1$ into the series: $\sum_{n=1}^{\infty} (-3)^n/(n\cdot 3^n) = \sum_{n=1}^{\infty} (-1)^n/n$.
- • This is the alternating harmonic series, which converges by AST. (Include $x = -1$).
- **Test the Right Endpoint x = 5:**
- • Substitute $x = 5$ into the series: $\sum_{n=1}^{\infty} 3^n/(n\cdot 3^n) = \sum_{n=1}^{\infty} 1/n$.
- • This is the harmonic series, which diverges. (Exclude $x = 5$).
- **Conclusion:**
- • The interval of convergence is **[−1, 5)**.

WHY OTHER OPTIONS ARE INCORRECT

Question 62 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) [−5, 3]

STEP-BY-STEP SOLUTION

- **Find the Radius of Convergence via the Ratio Test:**
- • $L = \lim_{n\to\infty} |[(x+1)^{n+1} / ((n+1)^2 \cdot 4^{n+1})] \cdot [(n^2\cdot 4^n)/(x+1)^n]|$
- • $= \lim_{n\to\infty} |x+1|/4 \cdot [n/(n+1)]^2 = |x+1|/4$.
- • $|x+1|/4 < 1 \Rightarrow |x+1| < 4$ (Radius $R = 4$).
- • $-4 < x+1 < 4 \Rightarrow -5 < x < 3$.
- **Test the Right Endpoint x = 3:**
- • $x = 3 \Rightarrow \sum 4^n/(n^2\cdot 4^n) = \sum 1/n^2$ (convergent p-series, $p = 2 > 1$). (Include $x = 3$).
- **Test the Left Endpoint x = −5:**
- • $x = -5 \Rightarrow \sum (-4)^n/(n^2\cdot 4^n) = \sum (-1)^n/n^2$ (absolutely convergent p-series). (Include $x = -5$).
- **Conclusion:**
- • The interval of convergence is the closed interval **[−5, 3]**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Fails to test endpoints, leaving the interval open.

✗ **(C)** Excludes $x = 3$, overlooking that $\sum 1/n^2$ converges.

Question 63 Analysis

★★★★ MANDATORY

Correct Answer: (C) (−1 / 2, 9 / 2)

STEP-BY-STEP SOLUTION

- **Identify as a Geometric Series:**
- • The series can be written as $\sum_{n=0}^{\infty} [(2x-4)/5]^n$.
- • Common ratio $r = (2x-4)/5$.
- **Convergence Condition for Geometric Series:**
- • $|r| < 1 \Rightarrow |(2x-4)/5| < 1 \Rightarrow |2x-4| < 5$.
- • $-5 < 2x-4 < 5$.
- • Add 4: $-1 < 2x < 9$.
- • Divide by 2: $-1/2 < x < 9/2$.
- **Check the Endpoints:**
- • At $x = -1/2$: $r = -1$, series is $\sum (-1)^n$, which diverges by oscillation.
- • At $x = 9/2$: $r = 1$, series is $\sum 1^n = \sum 1$, which diverges.
- • Geometric series *never* converge at their endpoints $|r| = 1$.
- **Conclusion:**
- • The interval of convergence is the open interval **(−1 / 2, 9 / 2)**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Includes the right endpoint.

✗ **(B)** Falsely includes both endpoints; geometric series diverge at $|r| = 1$.

✗ **(D)** Fails to solve for x, reporting the bound on $2x - 4$.

Question 64 Analysis

★★★★ MANDATORY

Correct Answer: (D) R = 0 (converges only at x = 3)

STEP-BY-STEP SOLUTION

- **Apply the Ratio Test:**
- • $L = \lim_{n\to\infty} \left| \left[(n+1)!\,(x-3)^{n+1} \right] / \left[n!\,(x-3)^{n} \right] \right|$
- • $= \lim_{n\to\infty} (n+1)\,|x-3|$.
- **Evaluate the Limit for $x \neq 3$:**
- • If $x \neq 3$, $|x-3| > 0$, so $\lim_{n\to\infty} (n+1)\,|x-3| = \infty > 1$.
- • Thus, the series diverges for every $x \neq 3$.
- **Evaluate at the Center x = 3:**
- • When x = 3, every term after n = 0 is 0, so the series trivially sums to 0! (1) = 1.
- **Conclusion:**
- • The series converges only at its center x = 3; therefore, the radius of convergence is **R = 0**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Mistakes the center c = 3 for the radius R.

✗ **(B)** Confuses n! in the numerator with n! in the denominator (which yields R = ∞).

✗ **(C)** Thinks all power series have radius at least 1.

Question 65 Analysis

★★★★ MANDATORY

Correct Answer: (A) (−∞, ∞)

STEP-BY-STEP SOLUTION

- **Apply the Ratio Test:**
- • $L = \lim_{n\to\infty} \left| \left[(x-1)^{n+1} / (n+1)! \right] \cdot \left[n! / (x-1)^{n} \right] \right|$
- • $= \lim_{n\to\infty} |x-1| / (n+1)$.
- **Evaluate the Limit for Any Fixed Real x:**
- • For any fixed real number x, $|x-1|$ is a constant.
- • As $n \to \infty$: $\lim |x-1| / (n+1) = 0$.
- **Conclusion:**
- • Since $L = 0 < 1$ for all real numbers x, the series converges absolutely everywhere.
- • Therefore, the radius is $R = \infty$ and the interval of convergence is **(−∞, ∞)**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Assumes the radius of convergence is R = 1.

✗ **(C)** Open interval of radius 1.

✗ **(D)** Confuses n! in the denominator with n! in the numerator.

Question 66 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) The series converges at x = 2 and diverges at x = 9.

STEP-BY-STEP SOLUTION

- **Determine the Center of the Power Series:**
- • The series is centered at c = 4.
- **Analyze Convergence at x = 7:**
- • Distance from center: $|7-4| = 3$.
- • Because the series converges at x = 7, the radius of convergence must satisfy: $R \geq 3$.
- • By the Power Series Convergence Theorem, the series converges absolutely for all x with $|x-4| < 3$ (i.e. on (1, 7)).
- **Analyze Divergence at x = 0:**
- • Distance from center: $|0-4| = 4$.
- • Because the series diverges at x = 0, the radius of convergence must satisfy: $R \leq 4$.
- • By theorem, the series diverges for all x with $|x-4| > 4$ (i.e. $x < 0$ or $x > 8$).
- **Test the Candidate Points:**
- • At x = 2: $|2-4| = 2$. Since $2 < 3 \leq R$, the series **must converge** at x = 2.
- • At x = 9: $|9-4| = 5$. Since $5 > 4 \geq R$, the series **must diverge** at x = 9.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Reverses the convergence and divergence classifications.

✗ **(C)** x = 9 is distance 5 from center, which is outside the maximum radius $R \leq 4$.

✗ **(D)** R could be any value between 3 and 4 (e.g. R = 3.5); it is not guaranteed to be exactly 3.

Question 67 Analysis

★★★★ MANDATORY

Correct Answer: (C) [3, 7)

STEP-BY-STEP SOLUTION

- **Synthesize Table Data:**
- • The interior of convergence is (3, 7) because the center is c = 5 and the radius is R = 2.
- • At the left endpoint x = 3: the series converges by the Alternating Series Test. Therefore, x = 3 is **included**: [3.
- • At the right endpoint x = 7: the series diverges by the p-series test (p = 1/2 ≤ 1). Therefore, x = 7 is **excluded**: 7).
- **Conclusion:**
- • The interval of convergence is **[3, 7)**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Leaves both endpoints open.
✗ **(B)** Includes 7 and excludes 3, reversing the table findings.
✗ **(D)** Includes 7 despite the table explicitly proving divergence.

Question 68 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) (−2, 2)

STEP-BY-STEP SOLUTION

- **Apply the Ratio Test:**
- • $L = \lim_{n\to\infty} \left| \left[x^{2n+2} / ((n+1)\cdot 4^{n+1}) \right] \cdot \left[(n \cdot 4^n) / x^{2n} \right] \right|$
- • $= \lim_{n\to\infty} \left[x^2 / 4 \right] \cdot \left[n / (n+1) \right] = x^2 / 4$.
- **Solve the Inequality L < 1:**
- • $x^2 / 4 < 1 \Rightarrow x^2 < 4 \Rightarrow |x| < 2 \Rightarrow -2 < x < 2$ (Radius R = 2).
- **Test the Endpoints x = ±2:**
- • Note that for both x = 2 and x = −2: $x^2 = 4$, so $x^{2n} = (x^2)^n = 4^n$.
- • At both x = 2 and x = −2, the series becomes: $\sum_{n=1}^{\infty} 4^n / (n \cdot 4^n) = \sum_{n=1}^{\infty} 1/n$.
- • This is the harmonic series, which diverges at both endpoints!
- **Conclusion:**
- • The interval of convergence is strictly open: **(−2, 2)**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses radius R = 4 instead of R = √4 = 2.
✗ **(B)** Includes both endpoints, failing to notice that $(-2)^{2n} = 4^n$ produces the divergent harmonic series.

Question 69 Analysis

★★★★ MANDATORY

Correct Answer: (A) R = 5

STEP-BY-STEP SOLUTION

- **Theorem on Term-by-Term Differentiation:**
- • If a power series $\sum a_n (x-c)^n$ has radius of convergence R, then the term-by-term differentiated series $\sum n\, a_n (x-c)^{n-1}$ has the **exact same radius of convergence R**.
- • The center c and radius R remain completely unchanged by differentiation (and integration).
- • (Note: Only the convergence behavior at the boundary endpoints may potentially change).
- **Conclusion:**
- • The radius of convergence of f'(x) is **R = 5**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Mistakenly subtracts 1 from the radius due to n − 1 power.
✗ **(C)** Squares the radius.
✗ **(D)** Falsely assumes differentiating factorials or terms collapses convergence.

Question 70 Analysis

★★★★ MANDATORY

Correct Answer: (B) (−4, −2]

STEP-BY-STEP SOLUTION

- **Find the Radius of Convergence via the Ratio Test:**
- • $L = \lim_{n\to\infty} \left| [(x+3)^{n+1}/(n+1)] \cdot [n/(x+3)^n] \right| = |x+3|$.
- • Set $L < 1$: $|x+3| < 1 \Rightarrow -1 < x+3 < 1 \Rightarrow -4 < x < -2$ (Radius $R = 1$).
- **Test the Right Endpoint x = −2:**
- • $x = -2 \Rightarrow x+3 = 1$.
- • Series: $\sum_{n=1}^{\infty} (-1)^n (1)^n / n = \sum_{n=1}^{\infty} (-1)^n / n$.
- • This is the alternating harmonic series, which converges by AST. (Include $x = -2$).
- **Test the Left Endpoint x = −4:**
- • $x = -4 \Rightarrow x+3 = -1$.
- • Series: $\sum_{n=1}^{\infty} (-1)^n (-1)^n / n = \sum_{n=1}^{\infty} (-1)^{(2n)} / n = \sum_{n=1}^{\infty} 1/n$.
- • This is the regular harmonic series, which diverges. (Exclude $x = -4$).
- **Conclusion:**
- • The interval of convergence is **(−4, −2]**.

WHY OTHER OPTIONS ARE INCORRECT

Question 71 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) $\sum_{n=0}^{\infty} [(-1)^n 3^n x^{(n+2)}] / n!$

STEP-BY-STEP SOLUTION

- **Recall the Maclaurin Series for e^u:**
- • $e^u = \sum_{n=0}^{\infty} u^n / n! = 1 + u + u^2/2! + u^3/3! + \dots$.
- **Substitute u = −3x:**
- • $e^{-3x} = \sum_{n=0}^{\infty} (-3x)^n / n! = \sum_{n=0}^{\infty} [(-1)^n 3^n x^n] / n!$.
- **Multiply by x^2:**
- • $x^2 e^{-3x} = x^2 \sum_{n=0}^{\infty} [(-1)^n 3^n x^n] / n! = \sum_{n=0}^{\infty} [(-1)^n 3^n x^{(n+2)}] / n!$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to multiply through by x^2, leaving power x^n.
✗ **(B)** Forgets the alternating sign $(-1)^n$ from $(-3x)^n$.
✗ **(D)** Incorrectly multiplies exponents, writing x^(2n) instead of x^(n+2).

Question 72 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) 4 / 15

STEP-BY-STEP SOLUTION

- **Recall the Maclaurin Series for sin(u):**
- • $\sin(u) = u - u^3/3! + u^5/5! - u^7/7! + \dots$.
- **Substitute u = 2x:**
- • The term containing u^5 is $+u^5/5! = (2x)^5/5!$.
- **Evaluate the Coefficient:**
- • $(2x)^5 = 32x^5$.
- • $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$.
- • Coefficient = 32 / 120.
- • Divide numerator and denominator by 8: 32 / 120 = **4 / 15**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error; the u^5 term in the sine series has a positive coefficient.
✗ **(B)** Forgets to divide by $5! = 120$.
✗ **(C)** Fails to raise 2 to the fifth power properly (using 16 instead of 32).

Question 73 Analysis

★★★★ MANDATORY

Correct Answer: (A) $\sum_{n=0}^{\infty} (-1)^n 4^n x^{(2n+1)}$ for $|x| < 1/2$

STEP-BY-STEP SOLUTION

- **Use the Geometric Series Formula:**
- • $1/(1-u) = \sum_{n=0}^{\infty} u^n$ for $|u| < 1$.
- • Rewrite the denominator: $1/(1+4x^2) = 1/[1-(-4x^2)]$.
- • Substitute $u = -4x^2$:
- • $1/(1+4x^2) = \sum_{n=0}^{\infty} (-4x^2)^n = \sum_{n=0}^{\infty} (-1)^n 4^n x^{(2n)}$.
- **Multiply by x:**
- • $x/(1+4x^2) = x \sum_{n=0}^{\infty} (-1)^n 4^n x^{(2n)} = \sum_{n=0}^{\infty} (-1)^n 4^n x^{(2n+1)}$.
- **Find the Convergence Interval:**
- • Convergence requires $|u| < 1 \Rightarrow |-4x^2| < 1 \Rightarrow 4x^2 < 1 \Rightarrow x^2 < 1/4 \Rightarrow |x| < 1/2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Omits the alternating sign $(-1)^n$ generated by the positive sign in $1+4x^2$.
✗ **(C)** Forgets to multiply through by x and calculates radius as 2.
✗ **(D)** Uses 2^n instead of 4^n.

Question 74 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $\sum_{n=0}^{\infty} [(-1)^n (x-\pi/2)^{(2n)}]/(2n)!$

STEP-BY-STEP SOLUTION

- **Evaluate Derivatives of sin x at $c = \pi/2$:**
- • $f(\pi/2) = \sin(\pi/2) = 1$.
- • $f'(\pi/2) = \cos(\pi/2) = 0$.
- • $f''(\pi/2) = -\sin(\pi/2) = -1$.
- • $f'''(\pi/2) = -\cos(\pi/2) = 0$.
- • $f^{(4)}(\pi/2) = \sin(\pi/2) = 1$.
- **Observe the Pattern:**
- • All odd-order derivatives vanish at $\pi/2$.
- • The even-order derivatives alternate: $1, -1, 1, -1, \ldots$.
- **Construct the Taylor Series:**
- • $P(x) = 1 - (x-\pi/2)^2/2! + (x-\pi/2)^4/4! - \ldots = \sum_{n=0}^{\infty} [(-1)^n (x-\pi/2)^{(2n)}]/(2n)!$.
- • (Note: This is because $\sin(x) = \cos(x-\pi/2)$).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses the odd-power Maclaurin series centered at 0

Question 75 Analysis

★★★★ MANDATORY

Correct Answer: (C) $1 - x^4/2 + x^8/24$

STEP-BY-STEP SOLUTION

- **Recall the Maclaurin Series for cos(u):**
- • $\cos(u) = 1 - u^2/2! + u^4/4! - u^6/6! + \ldots$.
- **Substitute $u = x^2$:**
- • $\cos(x^2) = 1 - (x^2)^2/2! + (x^2)^4/4! - \ldots$
- • $= 1 - x^4/2 + x^8/24$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses the Maclaurin series for sine instead of cosine.
✗ **(B)** Forgets to square x, leaving powers of x^2 and x^4.
✗ **(D)** Omits alternating signs, writing all addition.

Question 76 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) 1 / 2

STEP-BY-STEP SOLUTION

- **Recognize the Familiar Maclaurin Series:**
- • The Maclaurin series for cos(x) is: $\cos(x) = \sum_{n=0}^{\infty} [(-1)^n x^{(2n)}] / (2n)!$.
- **Evaluate at x = π / 3:**
- • The given numerical series matches this series with $x = \pi / 3$:
- • Sum = $\cos(\pi / 3)$.
- • From special right triangles: $\cos(\pi / 3) = \mathbf{1 / 2}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Takes only the first term at n = 0.
✗ **(B)** Evaluates $\sin(\pi/3) = \sqrt{3} / 2$ instead of $\cos(\pi/3)$.
✗ **(C)** Guesses 0 without recognizing the series.

Question 77 Analysis

★★★★ MANDATORY

Correct Answer: (A) f(x) = cos x

STEP-BY-STEP SOLUTION

- **Analyze the Non-Zero Coefficients:**
- • $a_0 = 1$
- • $a_2 = -1/2 = -1 / 2!$
- • $a_4 = 1/24 = 1 / 4!$
- • All odd coefficients are zero: $a_1 = a_3 = a_5 = 0$.
- **Match with Known Maclaurin Series:**
- • The series is: $1 - x^2 / 2! + x^4 / 4! - \ldots$.
- • This is precisely the Maclaurin series for **f(x) = cos x**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** sin x has only odd powers: $x - x^3/6 + \ldots$.
✗ **(C)** e^x has non-zero coefficients for every integer power of x.
✗ **(D)** $1 / (1 + x^2) = 1 - x^2 + x^4 - \ldots$ without factorials in the denominators.

Question 78 Analysis

★★★★ MANDATORY

Correct Answer: (B) e^2

STEP-BY-STEP SOLUTION

- **Recall the Maclaurin Series for the Exponential Function:**
- • $e^x = \sum_{n=0}^{\infty} x^n / n!$ for all real numbers x.
- **Substitute x = 2:**
- • Setting x = 2 gives: $\sum_{n=0}^{\infty} 2^n / n! = \mathbf{e^2}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Multiplies e by 2 instead of squaring e.
✗ **(C)** Divides e by 2.
✗ **(D)** Incorrectly treats the series as geometric with r = 2.

Question 79 Analysis

★★★★ MANDATORY

Correct Answer: (C) ln 2

STEP-BY-STEP SOLUTION

- **Recall the Maclaurin Series for ln(1 + x):**
- • $\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} x^n / n = x - x^2/2 + x^3/3 - x^4/4 + \ldots$ for $-1 < x \leq 1$.
- **Evaluate at the Convergent Endpoint x = 1:**
- • Substituting x = 1 yields:
- • $1 - 1/2 + 1/3 - 1/4 + \ldots = \ln(1+1) =$ **ln 2**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** ln(1/2) = –ln 2, which is negative (the sum is clearly positive).
✗ **(B)** Arbitrary estimate from partial sums.
✗ **(D)** π / 4 is the sum of the Leibniz series 1 – 1/3 + 1/5 – 1/7 + … .

Question 80 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) $1 + x - x^3/3$

STEP-BY-STEP SOLUTION

- **Write the Series Expansions up to Degree 3:**
- • $e^x = 1 + x + x^2/2 + x^3/6 + \ldots$
- • $\cos x = 1 - x^2/2 + \ldots$
- **Multiply the Polynomials Term by Term:**
- • Degree 0: (1)(1) = 1.
- • Degree 1: (x)(1) = x.
- • Degree 2: $(1)(-x^2/2) + (x^2/2)(1) = -x^2/2 + x^2/2 =$ **0** (the x^2 term cancels!).
- • Degree 3: $(x)(-x^2/2) + (x^3/6)(1) = -x^3/2 + x^3/6 = -3x^3/6 + x^3/6 = -2x^3/6 =$ **$-x^3/3$**.
- **Collect the First Three Non-Zero Terms:**
- • $f(x) = 1 + x - x^3/3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits the linear term x.
✗ **(B)** Assumes the x^2 term does not cancel.
✗ **(C)** Adds coefficients without multiplying polynomial terms.

Question 81 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 23 / 30

STEP-BY-STEP SOLUTION

- **Write the Maclaurin Series for e^{-x^2}:**
- • $e^u = 1 + u + u^2/2! + \ldots \Rightarrow e^{-x^2} = 1 - x^2 + x^4/2 - \ldots$.
- **Integrate Term-by-Term from 0 to 1:**
- • $\int_0^1 (1 - x^2 + x^4/2)\,dx = [x - x^3/3 + x^5/10]_0^1$.
- **Evaluate the Limits:**
- • At x = 1: 1 – 1/3 + 1/10.
- • Common denominator of 30: 30 / 30 – 10 / 30 + 3 / 30 = 23 / 30.
- • (Note: 23 / 30 ≈ 0.767, which closely approximates the true value $\text{erf}(1)\sqrt{\pi}/2 \approx 0.747$).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Omits the third term: 1 – 1/3 = 2/3 ≈ 0.667 or arithmetic error.
✗ **(C)** Forgets to divide by 5 when integrating $x^4/2$, using $x^4/4$ instead.
✗ **(D)** Sign error in integrating $-x^2$.

Question 82 Analysis

★★★★ MANDATORY

Correct Answer: (B) $\sum_{n=0}^{\infty} [(-1)^n x^{(2n+1)}] / [(2n+1)(2n+1)!]$

STEP-BY-STEP SOLUTION

- **Find the Maclaurin Series for sin(t) / t:**
- • $\sin(t) = \sum_{n=0}^{\infty} [(-1)^n t^{(2n+1)}] / (2n+1)! = t - t^3/3! + t^5/5! - \dots$.
- • Divide by t: $\sin(t)/t = \sum_{n=0}^{\infty} [(-1)^n t^{(2n)}] / (2n+1)! = 1 - t^2/6 + t^4/120 - \dots$.
- **Integrate Term-by-Term with Respect to t from 0 to x:**
- • $\int_0^x [(-1)^n t^{(2n)} / (2n+1)!] dt = [(-1)^n t^{(2n+1)} / [(2n+1)(2n+1)!]]_0^x$
- • $= \sum_{n=0}^{\infty} [(-1)^n x^{(2n+1)}] / [(2n+1)(2n+1)!]$.
- • The first terms are: $x - x^3/(3 \cdot 6) + x^5/(5 \cdot 120) = x - x^3/18 + x^5/600 - \dots$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Fails to integrate, giving the series for sin(x).
✗ **(C)** Series for cos(x).
✗ **(D)** Integrates with an incorrect power shift.

Question 83 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) $1/(1-x)^2$

STEP-BY-STEP SOLUTION

- **Differentiate the Known Geometric Series Term-by-Term:**
- • $1/(1-x) = (1-x)^{-1} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots$ for $|x| < 1$.
- **Differentiate the Left Side:**
- • $d/dx [(1-x)^{-1}] = -(1-x)^{-2} \cdot (-1) = 1/(1-x)^2$.
- **Differentiate the Right Side:**
- • $d/dx [\sum_{n=0}^{\infty} x^n] = \sum_{n=1}^{\infty} n x^{n-1} = 1 + 2x + 3x^2 + \dots$.
- **Equate Both Sides:**
- • $\sum_{n=1}^{\infty} n x^{n-1} = \mathbf{1/(1-x)^2}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Multiplying by x gives $\sum n x^n = x/(1-x)^2$.
✗ **(B)** Differentiating a second time yields $2/(1-x)^3$.
✗ **(D)** Integrating $1/(1-x)$ yields $-\ln(1-x)$.

Question 84 Analysis

★★★★ MANDATORY

Correct Answer: (D) 1 / 2

STEP-BY-STEP SOLUTION

- **Replace e^x with Its Maclaurin Series:**
- • $e^x = 1 + x + x^2/2! + x^3/3! + \dots = 1 + x + x^2/2 + x^3/6 + \dots$.
- **Substitute into the Numerator:**
- • $e^x - 1 - x = (1 + x + x^2/2 + x^3/6 + \dots) - 1 - x = x^2/2 + x^3/6 + \dots$.
- **Divide by x^2:**
- • $(e^x - 1 - x)/x^2 = 1/2 + x/6 + \dots$.
- **Evaluate the Limit as $x \to 0$:**
- • $\lim_{x \to 0} [1/2 + x/6 + \dots] = 1/2 + 0 = \mathbf{1/2}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Indeterminate 0/0 form resolved directly by Taylor series.
✗ **(B)** Applying L'Hôpital's Rule once gives $\lim (e^x - 1)/(2x)$ and stopping prematurely might suggest 1.
✗ **(C)** Guesses 0 because numerator evaluated at 0 is 0.

Question 85 Analysis

★★★★ MANDATORY

Correct Answer: (A) 1 / 6

STEP-BY-STEP SOLUTION

- **Use the Maclaurin Series for sin x:**
- • $\sin x = x - x^3/3! + x^5/5! - \ldots = x - x^3/6 + x^5/120 - \ldots$.
- **Simplify the Numerator:**
- • $x - \sin x = x - (x - x^3/6 + x^5/120 - \ldots) = x^3/6 - x^5/120 + \ldots$.
- **Divide by x^3:**
- • $(x - \sin x)/x^3 = 1/6 - x^2/120 + \ldots$.
- **Take the Limit as $x \to 0$:**
- • $\lim_{x\to 0}[1/6 - x^2/120 + \ldots] = 1/6 - 0 =$ **1 / 6**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sign error in distributing the negative sign over the sine expansion.
✗ **(C)** Confuses 3! = 6 with 3.
✗ **(D)** Falsely assumes the numerator cancels completely to 0.

Question 86 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $\pi = 4[1 - 1/3 + 1/5 - 1/7 + 1/9 - \ldots]$

STEP-BY-STEP SOLUTION

- **Integrate Term-by-Term:**
- • $\arctan(x) = \int_0^x 1/(1+t^2)\,dt = \int_0^x \sum_{n=0}^{\infty} (-1)^n t^{(2n)}\,dt$.
- • $= \sum_{n=0}^{\infty} [(-1)^n x^{(2n+1)}]/(2n+1) = x - x^3/3 + x^5/5 - x^7/7 + \ldots$.
- **Evaluate at x = 1:**
- • $\arctan(1) = 1 - 1/3 + 1/5 - 1/7 + \ldots$.
- • Since $\arctan(1) = \pi/4$:
- • $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \ldots$.
- **Multiply by 4:**
- • $\pi = 4[1 - 1/3 + 1/5 - 1/7 + 1/9 - \ldots]$ (The Gregory-Leibniz series for π).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Multiplies by 2 instead of 4.
✗ **(C)** Omits the alternating signs generated by $(-1)^n$.
✗ **(D)** Formula for ln 2, not π.

Question 87 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 1 / 110592

STEP-BY-STEP SOLUTION

- **Identify the First Omitted Term in the Integral:**
- • The third term of the integrand is $+x^8/24$ (n = 2).
- • Integrate this term from 0 to 1/2:
- • $\int_0^{1/2} (x^8/24)\,dx = [x^9/(24 \cdot 9)]_0^{(1/2)} = [x^9/216]_0^{(1/2)}$.
- **Substitute x = 1/2:**
- • $(1/2)^9 = 1/512$.
- • Error Bound = $(1/512)/216 = 1/(512 \cdot 216)$.
- **Compute the Product:**
- • $512 \cdot 216 = 110{,}592$.
- • Error Bound ≤ **1 / 110592** ≈ 9.04×10^{-6}.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Reports the magnitude of the last included term 1 / 160.
✗ **(B)** Multiplies by 108 instead of 216.
✗ **(D)** Forgets to integrate, using $(1/2)^8/24 = 1/6144$.

Question 88 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) 2

STEP-BY-STEP SOLUTION

- **Recall the Derivative of the Geometric Series:**
- • For $|x| < 1$: $\sum_{n=1}^{\infty} n x^{n-1} = 1/(1-x)^2$.
- **Multiply Both Sides by x:**
- • $x \sum_{n=1}^{\infty} n x^{n-1} = \sum_{n=1}^{\infty} n x^n = x/(1-x)^2$.
- **Substitute x = 1 / 2 (Since |1/2| < 1):**
- • $\sum_{n=1}^{\infty} n (1/2)^n = (1/2)/[1-1/2]^2$
- • $= (1/2)/(1/4) = (1/2) \cdot 4 = \mathbf{2}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Approximates by summing only the first two terms: 1/2 + 2/4 = 1, or 1 + 3/8.

✗ **(B)** Confuses with the sum of the simple geometric series $\sum (1/2)^n = 1$.

✗ **(C)** Forgets the 1/2 in the numerator: 1 / (1/4) = 4.

Question 89 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) –80 / 3

STEP-BY-STEP SOLUTION

- **Relate Series Coefficients to Derivatives:**
- • In the Maclaurin series, the coefficient of x^7 is: $a_7 = f^{(7)}(0)/7!$.
- **Find the Term in the Series with Power 7:**
- • Set the exponent $2n + 1 = 7 \Rightarrow 2n = 6 \Rightarrow n = 3$.
- **Calculate the Coefficient a_7 for n = 3:**
- • $a_7 = [(-1)^3]/[(2(3)+1) \cdot 3^3] = -1/[7 \cdot 27] = -1/189$.
- **Solve for f^(7)(0):**
- • $f^{(7)}(0) = 7! \cdot a_7 = 5040 \cdot (-1/189)$.
- • Divide 5040 and 189 by their greatest common factor 63:
- • 5040 / 63 = 80 and 189 / 63 = 3.
- • f^(7)(0) = **–80 / 3**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Fails to reduce the fraction –5040 / 189.

✗ **(C)** Reports the coefficient $a_7 = -1/189$ directly without multiplying by 7!

Question 90 Analysis

★★★★ MANDATORY

Correct Answer: (B) Approximation = 1 / 4 with error at most 1 / 96

STEP-BY-STEP SOLUTION

- **Find the Maclaurin Series for (1 – cos x) / x:**
- • $\cos x = 1 - x^2/2! + x^4/4! - \ldots = 1 - x^2/2 + x^4/24 - \ldots$.
- • $1 - \cos x = x^2/2 - x^4/24 + x^6/720 - \ldots$.
- • $(1 - \cos x)/x = x/2 - x^3/24 + x^5/720 - \ldots$.
- **Integrate Term-by-Term from 0 to 1:**
- • Term 1: $\int_0^1 (x/2)\,dx = [x^2/4]_0^1 = 1/4$.
- • Term 2: $\int_0^1 (-x^3/24)\,dx = [-x^4/96]_0^1 = -1/96$.
- **Apply the Alternating Series Error Bound:**
- • Because the integrated series alternates and satisfies the AST hypotheses, approximating the integral by its first term (1/4) produces an error bounded by the magnitude of the first omitted term:
- • Error ≤ **1 / 96**.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to integrate x / 2, using 1 / 2 instead of 1 / 4.

✗ **(C)** Uses the integrand coefficient 1/24 instead of the integrated term 1/96.

✗ **(D)** Uses sine series expansion instead of cosine.

FRQ 1: Maclaurin Series from a Differential Equation Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Compute f'(0) and f''(0) (2 Points)

- **Evaluate f'(0):**

$dy/dx = 2x + y$. At $x = 0$ with $y(0) = 1$:

$f'(0) = 2(0) + 1 = \mathbf{1}$.

- **Differentiate to find d^2y/dx^2:**

$d^2y/dx^2 = d/dx(2x + y) = 2 + dy/dx$.

At $x = 0$: $f''(0) = 2 + f'(0) = 2 + 1 = \mathbf{3}$.

Part (b): Third Derivative & Maclaurin Polynomial $P_3(x)$ (2 Points)

- **Differentiate to find d^3y/dx^3:**

$d^3y/dx^3 = d/dx(2 + dy/dx) = d^2y/dx^2$.

At $x = 0$: $f'''(0) = f''(0) = \mathbf{3}$.

- **Construct $P_3(x)$:**

$P_3(x) = f(0) + f'(0)x + [f''(0) / 2!]x^2 + [f'''(0) / 3!]x^3$

$= 1 + 1x + (3 / 2)x^2 + (3 / 6)x^3$

$= \mathbf{1 + x + (3 / 2)x^2 + (1 / 2)x^3}$.

Part (c): Approximation of f(0.2) (3 Points)

- **Substitute $x = 0.2 = 1/5$ into $P_3(x)$:**

$P_3(0.2) = 1 + 0.2 + (3 / 2)(0.2)^2 + (1 / 2)(0.2)^3$

$= 1 + 0.2 + 1.5(0.04) + 0.5(0.008)$

$= 1 + 0.2 + 0.06 + 0.004 = \mathbf{1.264}$ (or 158 / 125).

Part (d): Behavior at x = 0 (2 Points)

- **Analyze Critical Points:**

A relative extremum can only occur where the first derivative is zero or undefined.

Here, $f'(0) = 1 \neq 0$.

- **Conclusion:**

Because the derivative is positive and non-zero, f is strictly increasing at $x = 0$; thus, the graph has **neither a relative minimum nor a relative maximum** at $x = 0$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	$f'(0) = 1$: 1 pt $f''(0) = 3$: 1 pt	2
Part (b)	$f'''(0) = 3$: 1 pt Maclaurin polynomial $P_3(x) = 1 + x + (3/2)x^2 + (1/2)x^3$: 1 pt	2
Part (c)	Substitution of $x = 0.2$: 1 pt Evaluation $P_3(0.2) = 1.264$: 2 pts	3
Part (d)	Neither conclusion: 1 pt Reasoning using $f'(0) = 1 \neq 0$: 1 pt	2

FRQ 2: Power Series Interval of Convergence & Endpoint Analysis Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Radius of Convergence via Ratio Test (3 Points)

- **Set Up Ratio of Consecutive Terms:**

$L = \lim_{n\to\infty} \left| \left[(-1)^{n+1} (x-2)^{n+1} / ((n+1)\cdot 4^{n+1}) \right] \cdot \left[(n \cdot 4^n) / ((-1)^n (x-2)^n) \right] \right|$.

- **Simplify:**

$= \lim_{n\to\infty} \left[|x-2| / 4 \right] \cdot \left[n / (n+1) \right] = |x-2| / 4$.

- **Apply Condition $L < 1$:**

$|x-2| / 4 < 1 \Rightarrow |x-2| < 4$.

- **Conclusion:** The radius of convergence is **$R = 4$**.

Part (b): Test Endpoint $x = 6$ (2 Points)

- **Substitute $x = 6$:**

$x - 2 = 6 - 2 = 4$.

Series: $\sum_{n=1}^{\infty} \left[(-1)^n 4^n \right] / \left[n \cdot 4^n \right] = \sum_{n=1}^{\infty} (-1)^n / n$.

- **Convergence Check:**

This is the alternating harmonic series. Since $1/n > 0$, $1/(n+1) \le 1/n$, and $\lim_{n\to\infty} 1/n = 0$, the series **converges** by the Alternating Series Test.

Part (c): Test Endpoint $x = -2$ (2 Points)

- **Substitute $x = -2$:**

$x - 2 = -2 - 2 = -4$.

Series: $\sum_{n=1}^{\infty} \left[(-1)^n (-4)^n \right] / \left[n \cdot 4^n \right] = \sum_{n=1}^{\infty} \left[(-1)^n (-1)^n 4^n \right] / \left[n \cdot 4^n \right] = \sum_{n=1}^{\infty} (-1)^\wedge(2n) / n = \sum_{n=1}^{\infty} 1 / n$.

- **Convergence Check:**

This is the harmonic series (p-series with $p = 1 \le 1$), which **diverges**.

Part (d): Complete Interval of Convergence (2 Points)

- Combining the interior $(-2, 6)$ with endpoint results: $x = -2$ excluded, $x = 6$ included.
- Interval of Convergence: **$(-2, 6]$**.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	Sets up Ratio Test: 1 pt Calculates limit $L = \|x-2\| / 4$: 1 pt Radius $R = 4$: 1 pt	**3**
Part (b)	Evaluates at $x = 6$ to get $\sum (-1)^n / n$: 1 pt Identifies convergence via Alternating Series Test: 1 pt	**2**
Part (c)	Evaluates at $x = -2$ to get $\sum 1 / n$: 1 pt Identifies divergence via harmonic p-series ($p = 1$): 1 pt	**2**
Part (d)	Interval of convergence $(-2, 6]$: 2 pts	**2**

FRQ 3: Taylor Polynomial & Lagrange Error Bound with Derivative Table Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Third-Degree Maclaurin Polynomial $P_3(x)$ (3 Points)

- **Evaluate Coefficients:**

$a_0 = f(0) = 3$.

$a_1 = f'(0) = -2$.

$a_2 = f''(0) / 2! = 6 / 2 = 3$.

$a_3 = f'''(0) / 3! = -18 / 6 = -3$.

- **Write Polynomial:**

$P_3(x) = \mathbf{3 - 2x + 3x^2 - 3x^3}$.

Part (b): Approximation of f(0.5) (2 Points)

- **Substitute x = 0.5 = 1/2:**

$P_3(0.5) = 3 - 2(0.5) + 3(0.5)^2 - 3(0.5)^3$

$= 3 - 1 + 3(0.25) - 3(0.125) = 2 + 0.75 - 0.375 = \mathbf{2.375}$ (or 19 / 8).

Part (c): Lagrange Error Bound (2 Points)

- **State the Error Formula (n = 3, c = 0, x = 0.5):**

$|R_3(0.5)| \le [M / 4!] |0.5 - 0|^4$, where $M = \max |f^{(4)}(t)|$ on $[0, 0.5]$.

- **Evaluate:**

$M = 48$, $4! = 24$, and $(0.5)^4 = (1/2)^4 = 1/16$.

$|R_3(0.5)| \le [48 / 24] \cdot (1 / 16) = 2 \cdot (1 / 16) = 1 / 8 = 0.125$.

- **Conclusion:**

The approximation differs from f(0.5) by at most **1 / 8**.

Part (d): Behavior at x = 0 (2 Points)

- **Analyze Critical Points:**

Since $f'(0) = -2 \neq 0$, $x = 0$ is not a critical point, so f cannot have a relative minimum or maximum at $x = 0$.

- **Analyze Concavity:**

$f''(0) = 6 \neq 0$, so $x = 0$ is not an inflection point (the graph is concave up).

- **Conclusion:**

f has **neither** a relative extremum nor an inflection point at $x = 0$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	Coefficients evaluated correctly: 2 pts Polynomial $P_3(x) = 3 - 2x + 3x^2 - 3x^3$: 1 pt	**3**
Part (b)	Substitution and evaluation $P_3(0.5) = 2.375$: 2 pts	**2**
Part (c)	Lagrange setup: 1 pt Verification that bound ≤ 1/8: 1 pt	**2**
Part (d)	Neither conclusion: 1 pt Justification via $f'(0) = -2 \neq 0$: 1 pt	**2**

FRQ 4: Alternating Series Convergence & Remainder Estimation Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Alternating Series Test Verification (3 Points)

- **Identify Magnitude Terms:**

$b_n = 1/(2n-1)$ for $n \geq 1$.

- **Verify Condition 1 (Positive):**

For $n \geq 1$, $2n-1 \geq 1 > 0$, so $b_n > 0$.

- **Verify Condition 2 (Decreasing):**

$2(n+1)-1 = 2n+1 > 2n-1 \Longrightarrow 1/(2n+1) < 1/(2n-1)$, so $b_{n+1} \leq b_n$.

- **Verify Condition 3 (Limit Zero):**

$\lim_{n\to\infty} 1/(2n-1) = 0$.

- **Conclusion:** The series converges by the Alternating Series Test.

Part (b): Absolute vs. Conditional Convergence (2 Points)

- **Evaluate Series of Absolute Values:**

$\sum_{n=1}^{\infty} |(-1)^{n+1}/(2n-1)| = \sum_{n=1}^{\infty} 1/(2n-1)$.

- **Limit Comparison with Harmonic Series $\sum 1/n$:**

$\lim_{n\to\infty} [1/(2n-1)]/[1/n] = \lim_{n\to\infty} n/(2n-1) = 1/2$.

Since $0 < 1/2 < \infty$ and $\sum 1/n$ diverges, $\sum 1/(2n-1)$ diverges.

- **Conclusion:** The series converges **conditionally**.

Part (c): Third Partial Sum Error & Direction (2 Points)

- **Calculate Error Bound:**

$|S - S_3| \leq b_4 = 1/[2(4)-1] = 1/7 \approx 0.1429 < 0.15$.

- **Over/Underestimate Direction:**

$S_3 = 1 - 1/3 + 1/5$ ends on a positive term, and the next omitted term is $-1/7 < 0$.

Therefore, S_3 is an **overestimate** of the true sum S.

Part (d): Minimum Terms for Error < 0.01 (2 Points)

- **Set Up Inequality:**

$|R_N| \leq b_{N+1} = 1/[2(N+1)-1] = 1/(2N+1)$.

We require: $1/(2N+1) < 0.01 = 1/100 \Longrightarrow 2N+1 > 100 \Longrightarrow 2N > 99 \Longrightarrow N > 49.5$.

- **Smallest Integer: N = 50**.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	Verifies $b_n > 0$: 1 pt Verifies $b_{n+1} \leq b_n$: 1 pt Verifies $\lim b_n = 0$: 1 pt	3
Part (b)	Examines absolute series $\sum 1/(2n-1)$: 1 pt Proves conditional convergence via LCT with $\sum 1/n$: 1 pt	2
Part (c)	Error bound $\lvert R_3\rvert \leq b_4 = 1/7 \approx 0.143 < 0.15$: 1 pt Identifies S_3 as overestimate: 1 pt	2
Part (d)	Sets $1/(2N+1) < 0.01$: 1 pt Smallest integer $N = 50$: 1 pt	2

FRQ 5: Term-by-Term Integration of a Maclaurin Series Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Maclaurin Series for f(x) (3 Points)

- **Expand e^{-x}:**

$e^{-x} = 1 - x + x^2/2! - x^3/3! + x^4/4! - \dots$.

- **Subtract from 1:**

$1 - e^{-x} = x - x^2/2 + x^3/6 - x^4/24 + \dots$.

- **Divide by x:**

$f(x) = \mathbf{1 - x/2 + x^2/6 - x^3/24 + \dots}$

- **General Term:** $\sum_{n=0}^{\infty} [(-1)^n x^n]/(n+1)!$.

Part (b): Maclaurin Series for G(x) (2 Points)

- **Integrate Term-by-Term from 0 to x:**

$G(x) = \int_0^x [1 - t/2 + t^2/6 - t^3/24 + \dots]\,dt$

$= [t - t^2/4 + t^3/18 - t^4/96 + \dots]_0^x$

$= \mathbf{x - x^2/4 + x^3/18 - x^4/96 + \dots}$.

Part (c): Approximation of G(1) (2 Points)

- **Use First Two Terms at x = 1:**

$G(1) \approx 1 - (1)^2/4 = 1 - 1/4 = \mathbf{3/4}$ (or 0.75).

Part (d): Error Bound Justification (2 Points)

- **Check Alternating Series Criteria:**

At x = 1, G(1) is an alternating series whose terms strictly decrease in magnitude toward 0.

- **Apply Alternating Series Error Bound:**

The error in approximating by the first two terms is bounded by the magnitude of the third term:

$|G(1) - 3/4| \le (1)^3/18 = \mathbf{1/18}$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	Expansion of $1 - e^{-x}$: 1 pt Division by x yielding $1 - x/2 + x^2/6 - x^3/24$: 1 pt General term: 1 pt	**3**
Part (b)	Term-by-term integration: 1 pt Series $G(x) = x - x^2/4 + x^3/18 - x^4/96$: 1 pt	**2**
Part (c)	Evaluation $G(1) \approx 1 - 1/4 = 3/4$: 2 pts	**2**
Part (d)	Applies Alternating Series Error Bound: 1 pt Shows error $\le 1/18$: 1 pt	**2**

FRQ 6: Geometric Power Series, Derivative Series & Numerical Sums Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Maclaurin Series for g(x) (2 Points)

- **Geometric Series Formula:**

$1/(1-u) = \sum_{n=0}^{\infty} u^n$ for $|u| < 1$.

- **Substitute $u = -3x$:**

$g(x) = 1/[1-(-3x)] = \sum_{n=0}^{\infty} (-1)^n 3^n x^n$.

- **Interval of Convergence:**

$|-3x| < 1 \Rightarrow 3|x| < 1 \Rightarrow |x| < 1/3 \Rightarrow (-1/3, 1/3)$.

Part (b): Differentiate the Series for g'(x) (3 Points)

- **Differentiate Term-by-Term:**

$d/dx\left[\sum_{n=0}^{\infty} (-1)^n 3^n x^n\right] = \sum_{n=1}^{\infty} (-1)^n n 3^n x^{n-1}$.

- This matches $g'(x) = d/dx\left[(1+3x)^{-1}\right] = -3(1+3x)^{-2} = -3/(1+3x)^2$.

Part (c): Series for $h(x) = x/(1+3x)^2$ (2 Points)

- Notice that $h(x) = -(x/3)\, g'(x)$:

$h(x) = -(x/3) \sum_{n=1}^{\infty} (-1)^n n 3^n x^{n-1} = \sum_{n=1}^{\infty} (-1)^{n+1} n 3^{n-1} x^n$.

Part (d): Exact Numerical Sum (2 Points)

- Observe the series to sum: $S = \sum_{n=1}^{\infty} n(-1)^{n+1}(3/4)^n$.
- Rewrite using h(x): Note that $(-1)^{n+1} n (3/4)^n = 3\left[(-1)^{n+1} n 3^{n-1} (1/4)^n\right] = 3\, h(1/4)$.
- Since $x = 1/4$ is inside the interval of convergence $(-1/3, 1/3)$:

$h(1/4) = (1/4)/[1+3(1/4)]^2 = (1/4)/(7/4)^2 = (1/4)/(49/16) = (1/4)\cdot(16/49) = 4/49$.

- Thus: Sum $= 3 \cdot (4/49) =$ **12 / 49**.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	Series $\sum (-1)^n 3^n x^n$: 1 pt Interval $(-1/3, 1/3)$: 1 pt	2
Part (b)	Term-by-term differentiation: 2 pts Series $\sum_{n=1}^{\infty} (-1)^n n 3^n x^{n-1}$: 1 pt	3
Part (c)	Multiplies by $-x/3$: 1 pt Series representation for h(x): 1 pt	2
Part (d)	Evaluates at appropriate input: 1 pt Exact sum = 12 / 49: 1 pt	2

FRQ 7: Taylor Series from Function Values & Derivative Table Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Fourth-Degree Taylor Polynomial $P_4(x)$ (3 Points)

- **Calculate Coefficients:**

$a_0 = f(3) = 2$.

$a_1 = f'(3) = -4$.

$a_2 = f''(3) / 2! = 10 / 2 = 5$.

$a_3 = f'''(3) / 3! = -36 / 6 = -6$.

$a_4 = f^{(4)}(3) / 4! = 144 / 24 = 6$.

- **Form Polynomial:**

$P_4(x) = \mathbf{2 - 4(x - 3) + 5(x - 3)^2 - 6(x - 3)^3 + 6(x - 3)^4}$.

Part (b): Second-Degree Taylor Polynomial for $f'(x)$ (2 Points)

- **Differentiate $P_3(x)$:**

$\frac{d}{dx}[\,2 - 4(x - 3) + 5(x - 3)^2 - 6(x - 3)^3\,]$

$= \mathbf{-4 + 10(x - 3) - 18(x - 3)^2}$.

Part (c): Approximation of $f(3.1)$ (2 Points)

- **Evaluate at $x - 3 = 0.1$:**

$P_4(3.1) = 2 - 4(0.1) + 5(0.1)^2 - 6(0.1)^3 + 6(0.1)^4$

$= 2 - 0.4 + 5(0.01) - 6(0.001) + 6(0.0001)$

$= 1.6 + 0.05 - 0.006 + 0.0006 = \mathbf{1.6446}$.

Part (d): Evaluate the Limit (2 Points)

- **Substitute Taylor Expansion into the Limit:**

$f(x) = 2 - 4(x - 3) + 5(x - 3)^2 + \dots$.

$f(x) - 2 + 4(x - 3) = 5(x - 3)^2 - 6(x - 3)^3 + \dots$.

- **Divide by $(x - 3)^2$:**

$\lim_{x \to 3}[\,5 - 6(x - 3) + \dots\,] = 5 - 0 = \mathbf{5}$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	Calculates coefficients: 2 pts Polynomial $P_4(x)$: 1 pt	**3**
Part (b)	Differentiates P_3 or evaluates directly: 1 pt Polynomial for $f'(x) = -4 + 10(x - 3) - 18(x - 3)^2$: 1 pt	**2**
Part (c)	Substitution of $x = 3.1$: 1 pt Evaluation $P_4(3.1) = 1.6446$: 1 pt	**2**
Part (d)	Applies series expansion or L'Hôpital's: 1 pt Limit = 5: 1 pt	**2**

FRQ 8: Ratio Test, Radius of Convergence & Function Identification Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Radius and Interval of Convergence (3 Points)

- **Apply the Ratio Test:**

$L = \lim_{n\to\infty} \left| \left[(-1)^{n+1} x^{2n+2} / 9^{n+1} \right] \cdot \left[9^n / ((-1)^n x^{2n}) \right] \right| = x^2 / 9.$

- **Convergence Inequality:**

$x^2 / 9 < 1 \Rightarrow x^2 < 9 \Rightarrow |x| < 3.$

- Radius $R = 3$.
- **Endpoints $x = \pm 3$:**

$x^2 = 9 \Rightarrow \sum (-1)^n (9)^n / 9^n = \sum (-1)^n$, which diverges by oscillation.

- Interval of Convergence: **$(-3, 3)$**.

Part (b): Closed-Form Function Representation (2 Points)

- **Identify as a Geometric Series:**

$f(x) = \sum_{n=0}^{\infty} \left[-x^2 / 9 \right]^n.$

- First term $a = 1$, common ratio $r = -x^2 / 9$.
- Sum $= a / (1 - r) = 1 / \left[1 - (-x^2 / 9) \right] = 1 / \left[(9 + x^2) / 9 \right] =$ **$9 / (9 + x^2)$**.

Part (c): Value of f(√3) (2 Points)

- Since $\sqrt{3}$ is inside the interval $(-3, 3)$:

$f(\sqrt{3}) = 9 / \left[9 + (\sqrt{3})^2 \right] = 9 / \left[9 + 3 \right] = 9 / 12 =$ **$3 / 4$**.

Part (d): Antiderivative Power Series (2 Points)

- **Integrate Term-by-Term:**

$\int f(x)\, dx = C + \sum_{n=0}^{\infty} \left[(-1)^n x^{2n+1} \right] / \left[(2n + 1) \cdot 9^n \right].$

- (Note: This represents $3 \arctan(x / 3) + C$).

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	Ratio Test setup: 1 pt Radius $R = 3$: 1 pt Interval $(-3, 3)$: 1 pt	3
Part (b)	Identifies geometric series: 1 pt Function $f(x) = 9 / (9 + x^2)$: 1 pt	2
Part (c)	Evaluation $f(\sqrt{3}) = 3 / 4$: 2 pts	2
Part (d)	Term-by-term integration: 1 pt Series representation with constant C: 1 pt	2

FRQ 9: Applied Convergence Modeling — Medical Drug Accumulation Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Drug Amounts After 2nd and 3rd Injections (2 Points)

- **Day 1 (Immediate):** $D_1 = 150$ mg.
- **Day 2 (Immediate):**

$D_2 = 150 + 0.4(D_1) = 150 + 0.4(150) = 150 + 60 =$ **210 mg**.

- **Day 3 (Immediate):**

$D_3 = 150 + 0.4(D_2) = 150 + 0.4(210) = 150 + 84 =$ **234 mg**.

Part (b): Closed-Form Formula for D_n (3 Points)

- **Express D_n as a Geometric Partial Sum:**

$D_n = 150 + 150(0.4) + 150(0.4)^2 + \ldots + 150(0.4)^{n-1}$.

- **Use the Finite Geometric Sum Formula:**

$D_n = 150\,[\,1 - (0.4)^n\,] / [\,1 - 0.4\,] = 150\,[\,1 - (0.4)^n\,] / 0.6$

$=$ **$250\,[\,1 - (0.4)^n\,]$ mg**.

Part (c): Long-Term Maximum Drug Level (2 Points)

- **Take the Limit as $n \to \infty$:**

$\lim_{n\to\infty} D_n = \lim_{n\to\infty} 250\,[\,1 - (0.4)^n\,]$.

Since $|0.4| < 1$, $\lim_{n\to\infty} (0.4)^n = 0$.

- Limiting Maximum $= 250(1 - 0) =$ **250 mg**.

Part (d): Long-Term Minimum Drug Level (2 Points)

- **Compute Residual Level Before Next Dose:**

Just before an injection, 40% of the long-term peak remains:

Limiting Minimum $= 0.4 \cdot (250) =$ **100 mg** (or $250 - 150 = 100$ mg).

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	Second injection $D_2 = 210$ mg: 1 pt Third injection $D_3 = 234$ mg: 1 pt	2
Part (b)	Geometric partial sum setup: 2 pts Closed-form expression D_n: 1 pt	3
Part (c)	Infinite geometric series sum: 1 pt Limiting maximum = 250 mg: 1 pt	2
Part (d)	Calculates 40% of limiting maximum: 1 pt Limiting minimum = 100 mg: 1 pt	2

FRQ 10: Comprehensive Unit 10 Capstone Synthesis Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Evaluate f'(0) and f'''(0) (2 Points)

- **Relate Coefficients to Derivatives:**

The general Maclaurin series is $f(x) = \sum [f^{(k)}(0) / k!] x^k$.

- **Linear Term (k = 1):**

Coefficient of x is $1 \Rightarrow f'(0) / 1! = 1 \Rightarrow f'(0) = \mathbf{1}$.

- **Cubic Term (k = 3):**

Coefficient of x^3 is $-1 / 12$.

$f'''(0) / 3! = -1 / 12 \Rightarrow f'''(0) / 6 = -1 / 12 \Rightarrow f'''(0) = -6 / 12 = \mathbf{-1 / 2}$.

Part (b): Radius of Convergence via Ratio Test (3 Points)

- **Set Up Ratio of Successive Terms:**

$| a_{n+1} / a_n | = [|x|^{2n+3} / ((2n+3)! \cdot 2^{n+1})] \cdot [((2n+1)! \cdot 2^n) / |x|^{2n+1}]$

$= [x^2 / 2] \cdot [(2n+1)! / (2n+3)!] = [x^2 / 2] \cdot [1 / ((2n+3)(2n+2))]$.

- **Take the Limit as $n \to \infty$:**

$L = \lim_{n\to\infty} [x^2 / 2] \cdot [1 / ((2n+3)(2n+2))] = 0$ for any fixed real number x.

- **Conclusion:**

Since $L = 0 < 1$ for all x, the series converges everywhere: $\mathbf{R = \infty}$.

Part (c): Error Bound Verification at x = 1 (2 Points)

- **Analyze the Alternating Series at x = 1:**

$f(1) = 1 - 1/12 + 1/480 - \ldots$ is an alternating series satisfying the AST criteria.

- **Identify First Omitted Term:**

$P_3(1) = 1 - 1/12$.

The first omitted term is the term for n = 2:

$b_2 = 1^5 / [(2(2)+1)! \cdot 2^2] = 1 / [5! \cdot 4] = 1 / [120 \cdot 4] = 1 / 480$.

- **Compare with 1 / 400:**

$|f(1) - P_3(1)| \le 1 / 480 < 1 / 400$ (since $480 > 400$).

Part (d): Evaluate the Limit (2 Points)

- **Substitute the Maclaurin Series:**

$f(x) = x - x^3 / 12 + x^5 / 480 - \ldots$.

$f(x) - x = -x^3 / 12 + x^5 / 480 - \ldots$.

- **Divide by x^3:**

$[f(x) - x] / x^3 = -1 / 12 + x^2 / 480 - \ldots$.

- **Take the Limit as $x \to 0$:**

$\lim_{x\to 0} [-1 / 12 + x^2 / 480 - \ldots] = \mathbf{-1 / 12}$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	$f'(0) = 1$: 1 pt $f'''(0) = -1/2$: 1 pt	2
Part (b)	Ratio Test setup: 1 pt Evaluation $L = 0$: 1 pt Radius $R = \infty$: 1 pt	3
Part (c)	Identifies first omitted term: 1 pt Shows $b_3 = 1/480 < 1/400$: 1 pt	2
Part (d)	Series substitution: 1 pt Limit $= -1/12$: 1 pt	2

Essential Unit 10 Rules

Core convergence tests, Taylor and Maclaurin expansions, and error bound theorems required for AP Calculus BC.

- **Convergence Tests for Infinite Series:** • **nth-Term Test for Divergence:** If $\lim_{n\to\infty} a_n \neq 0$ (or does not exist), $\sum a_n$ diverges. If $\lim a_n = 0$, the test is strictly inconclusive! • **Geometric Series:** $\sum_{n=0}^{\infty} a r^n = a / (1 - r)$ if $|r| < 1$. Diverges if $|r| \geq 1$. • **p-Series Test:** $\sum 1 / n^p$ converges if $p > 1$; diverges if $p \leq 1$ (Harmonic series when $p = 1$). • **Integral Test:** If $f(x)$ is continuous, positive, and decreasing on $[1, \infty)$, $\sum_{n=1}^{\infty} f(n)$ and $\int_1^{\infty} f(x)\,dx$ share the same convergence behavior. Remainder: $R_N \leq \int_N^{\infty} f(x)\,dx$. • **Direct Comparison Test (DCT):** If $0 \leq a_n \leq b_n$ and $\sum b_n$ converges $\Rightarrow \sum a_n$ converges. If $a_n \geq c_n \geq 0$ and $\sum c_n$ diverges $\Rightarrow \sum a_n$ diverges. • **Limit Comparison Test (LCT):** If $\lim_{n\to\infty} (a_n / b_n) = L$ with $0 < L < \infty$, then both series share the exact same convergence behavior. • **Alternating Series Test (AST):** $\sum (-1)^{n+1} b_n$ converges if $b_n > 0$, $b_{n+1} \leq b_n$, and $\lim_{n\to\infty} b_n = 0$. • **Alternating Series Error Bound:** $|S - S_N| \leq b_{N+1}$ (error is bounded by the magnitude of the first omitted term). • **Ratio Test:** $L = \lim_{n\to\infty} |a_{n+1} / a_n|$. Converges absolutely if $L < 1$; diverges if $L > 1$; inconclusive if $L = 1$.
- **Taylor and Maclaurin Series:** • Taylor Polynomial Centered at c: $P_n(x) = \sum_{k=0}^{n} [\, f^{(k)}(c) / k! \,] (x - c)^k$. • Maclaurin Polynomial ($c = 0$): $P_n(x) = \sum_{k=0}^{n} [\, f^{(k)}(0) / k! \,] x^k$. • Taylor Series Coefficients: $a_k = f^{(k)}(c) / k! \Rightarrow f^{(k)}(c) = k! \cdot a_k$.
- **Common Maclaurin Series to Memorize:** • $1 / (1 - x) = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \ldots$ for $|x| < 1$. • $e^x = \sum_{n=0}^{\infty} x^n / n! = 1 + x + x^2/2! + x^3/3! + \ldots$ for $(-\infty, \infty)$. • $\sin x = \sum_{n=0}^{\infty} [\, (-1)^n x^{(2n+1)} \,] / (2n + 1)! = x - x^3/3! + x^5/5! - \ldots$ for $(-\infty, \infty)$. • $\cos x = \sum_{n=0}^{\infty} [\, (-1)^n x^{(2n)} \,] / (2n)! = 1 - x^2/2! + x^4/4! - \ldots$ for $(-\infty, \infty)$. • $\ln(1 + x) = \sum_{n=1}^{\infty} [\, (-1)^{n-1} x^n \,] / n = x - x^2/2 + x^3/3 - \ldots$ for $(-1, 1]$. • $\arctan x = \sum_{n=0}^{\infty} [\, (-1)^n x^{(2n+1)} \,] / (2n + 1) = x - x^3/3 + x^5/5 - \ldots$ for $[-1, 1]$.
- **Lagrange Error Bound (Taylor's Theorem Remainder):** • $|R_n(x)| = |f(x) - P_n(x)| \leq [\, M / (n + 1)! \,] |x - c|^{(n+1)}$, where $M = \max |f^{(n+1)}(t)|$ for t between c and x.

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Thank You, Students!

Mastery Comes Through Deliberate Practice

Congratulations on completing this comprehensive 100-Question Priority-Starred Master Repository for Unit 10! Infinite sequences and series represent the ultimate summit of AP Calculus BC, unlocking power series representations of transcendental functions and rigorous error bounds. Review your steps, reflect on the distractor rationales, and approach your AP Calculus BC examination with absolute mastery, confidence, and precision!

TARGET SCORE: 5 · AP CALCULUS BC